MODULE 1

HEALTH AND SAFETY

UNIT 1.1: HEALTH AND SAFETY

Hygiene

Learners should be able to:

describe a hygienic environment.

practise hygiene within different environments.

discuss illnesses associated with unhygienic environment.

A hygienic environment is one that is free from the potential of spreading diseases. Examples of these environments include homes, schools and public places.

Ways of practicing hygiene in different environments

Home

It is essential to keep clothes clean and free from bad smells. Always wear clean clothes everyday. Beds we sleep in should be kept clean so that germs and biting insects that can cause rashes and other skin related infections. Food, dishes and pots hygiene is also essential because it can prevent food poisoning and diarrhea. It important to handle and cook food properly.

Schools

Schools like home should be well ventilated by keeping windows and doors open at most times to allow air to circulate freely. In schools toilets should be cleaned thoroughly with disinfectants and kept dry at all times. Toilet seats should also be kept closed to prevent flies from carrying disease our food. Litter of all sorts should be kept in closed rubbish bin or properly disposed.

Public places

Public places include parks, hospitals, bus ranks and others. We should keep these places clean. People should not urinate, wash clothes, defecate or litter into rivers, ponds, streams, culverts and bus ranks. These unhealthy habits can assist with the spread of germs and diseases such as tuberculosis, cholera, bilharzias and cancer. Poor disposal of certain chemicals can also be a cause of skin cancer.

Illnesses associated with unhygienic environments

Malaria, cholera, tuberculosis, bilharzias, dysentery, skin cancer and scabies.

Suggested Teaching and Learning Activities

- Pick litter at home and at school and dispose them properly.
- Make a poster to illustrate the types of diseases that are related to water.
- Write short articles on a hygienic environment.
- Carry out an awareness campaign against unhygienic environment in the school or community.

Support Materials

⇒ Posters, refuse bags, rubbish bins

⇒ Home utensils e.g. plates, pots etc.

⇒ Detergents, disinfectants, soaps etc.

Glossary

Bilharzias	-	a chronic illness that results from infection of the blood with a parasitic flatworm.
Cholera	-	an acute and often fatal intestinal disease that produces severe diarrhea, vomiting and gastric pain.
Diarrhea	-	frequent and excessive discharging of the bowels producing abnormally thin watery feces.
Dysentery	-	the disease of the lower intestine caused by infection with bacteria or parasites.
Environment	-	the conditions that surround people and affect the way they live.
Food Poisoning	-	an acute inflammation of the stomach and intestines caused by eating food contaminated with toxic or poisonous substances.
Malaria	-	an infectious disease caused by a parasite that is transmitted by bite of infected mosquitoes.
Pollution	-	anything that causes harm to the environment.
Refuse	-	things thrown away as being of no value or use, especially household garbage.
Scabies	-	a contagious disease marked by intense itching and inflammation.
Tuberculosis	-	an infectious disease that causes small rounded swellings to form on mucous membranes.
Ventilation	-	the movement or circulation of fresh air.

Assessment

- Identify different types of environments.
- List ways of keeping the home, schools and public places clean.
- Explain the effects of littering on the environment.
- Demonstrate ways of keeping self, home and school clean.
- List ways of avoiding food poisoning and diarrhea.
- Explain a clean environment.
- Explain how water related diseases could be prevented.

Safety and first aid

Learners should be able to:

follow simple instructions to minimise risks.
label tools, equipment, instruments and implements for storage.
work safely with a range of materials, tools, equipment, instruments and implements.
explain first aid.
discuss the importance of first aid.

The teacher should explain what safety is all about. Adequate examples of safety, including safe and unsafe practices should be given to the learners.

Examples of hazards

Littering, land, water and air pollution, storage of medicine and chemicals safely, use and disposal of household chemicals properly. Explain to learners the dangers of hazards in the environment and their prevention. Allow learners to carry out investigations on common hazards and report to the class.

Safety
Encourage learners to work safely at all times and report to the teacher any problems they encounter when working with materials, tools and equipment/instruments. It is important that the teacher demonstrates to the learners the proper ways of handling materials, tools and equipment.

Safety charts and symbols
Learners should be exposed to the safety regulations that are applicable to Creative and Performing Arts. Safety charts, preferably with comprehensive illustrations, should be displayed at strategic places in the classroom/workshop. The teacher should also explain the different symbols or warning signs that are used to alert people against hazards or dangers. Explain to the learners that warning signs use shapes and colours to convey warning against hazards and unsafe practice . Provide as many examples as possible.

Suggested Teaching and Learning Activities
- Allow learners to identify common hazards in their environment.
- Expose learners to some of the hazards of working with materials, tools, equipment and instruments.
- Allow learners to interpret the different warning signs.

Support Materials
- ⇒ Magazines showing common hazards in their environment.
- ⇒ Illustrated safety charts and warning signs.
- ⇒ Charts illustrating the common hazards of working with different materials, tools and equipment.

Glossary

Safety	-	the condition of being safe from undergoing or causing hurt, injury or loss
Hazard	-	risk or danger
Pollution	-	the state of making (air, water and earth) dirty or harmful to people, animals and plants
Chemical	-	relating to, used in or produced by chemistry
Household	-	domestic, family and home

Assessment
- List some safety precautions of using hand tools in the school workshop.
- Explain the dangers associated with using sports equipment?
- Identify and explain different warning signs.
- Explain some of the dangers of working with materials.

UNIT 1.2 FOOD AND NUTRITION

Food

Learners should be able to:

identify different foodstuffs.
describe food.
classify foods according to their groups and functions.
prepare a balanced breakfast.

Food and their functions

Body building foods: Proteins

Proteins are found in:

- food from animals such as meat, fish, cheese, milk and eggs.
- food from plants such as beans, nuts and wheat.

Energy giving foods: Carbohydrates, fats and oils

Examples include cereals, starchy foods, bread, maize, millet, sorghum, margarine, butter, lard, nuts and vegetable oils

Protective foods: Vitamins and minerals

Examples include milk, internal organs of animals (offal), fruit, vegetables, beans, cereals, and eggs.

Breakfast

Breakfast is normally a morning meal. It should consist of fruits or fruit juices, cereals, proteins and beverages.

Suggested Teaching and Learning Activities

- Show a selection of foodstuffs and pupils should arrange them in appropriate groups
- Divide pupils into groups of four
- Each group should plan a breakfast
- Each pupil should prepare a dish for his breakfast
- Serve the breakfast and discuss each breakfast

Support Materials

⇒ Posters on food groups, foods, refuse bags, rubbish bins
⇒ Home utensils - crockery such as plates, pots and cutlery e.g. spoons, knives, forks etc.
⇒ Cleaning agents such as detergents, disinfectants, soaps, etc.
⇒ Cleaning equipment such as brooms, brushes, dustpans, etc.

Glossary

Breakfast	-	The first meal of the day that is eaten in the morning.
Food	-	Any substance which contains nutrients required by our bodies to enable them to perform their functions effectively.
Dish	-	Any prepared food
Nutrients	-	Any substance found in food which nourishes our bodies.

Assessment

- Give one function of the following: Body building foods, energy giving foods and protective foods
- List two examples of foods for each food group.
- Plan and prepare a balanced breakfast.

UNIT 1.3 PERSONAL FITNESS

Exercise

Learners should be able to:

identify different types of exercise.
explain high and low intensity exercises.
perform high and low intensity exercises.

describe the effects of exercise on the body.
design a personal training Programme to improve health.
follow individual training programmes to keep fit.

The teacher should discuss the importance of good health practices. The discussion should include among others the importance of participating in physical activity exercise and fitness. The different types of exercises that vary from stretching, agility, coordination, speed, balance, strength, flexibility and endurance should also be discussed. The effects of exercise on the body as well as performing activities that contribute to good health should be emphasised. There is also need to explain the importance of safety when exercising so as to reduce injuries. Learners should be taught about safety precautions to consider when one is injured or during practical activities. The body reacts differently to exercise such as sweating, body temperature going up and increased pulse. These should be felt by learners themselves during practical activities. After that the teacher should explain how exercising regularly can improve individual's health e.g. developing strong lungs, heart and muscles. Improved concentration and toned muscles.

Suggested Teaching and Learning Activities

- Discuss the importance of good health.
- Ask pupils to bring pictures of people showing good health.
- Identify exercises for strength, power, speed, agility, co-ordination and flexibility and endurance that can be performed as warm up and cool down.
- Perform selected exercises to enhance fitness and good health during warm up and cool down.
- Describe how exercise can affect the body.

All these activities should be to the level of the learners as well as the content coverage in terms of depth, relevance and difficulty.

Support Materials

⇒ Magazines and posters of people who look healthy
⇒ Skipping ropes, hoops, tennis and plastic balls
⇒ Radio cassette, recorded tapes and music tapes

Glossary

Exercise	-	any physical activity
Health	-	a state of complete absence of disease
Agility	-	the ability to change direction accurately and quickly while moving rapidly
Co-ordination	-	the integration of separate abilities into the smooth execution of a task
Speed	-	the quickness with which one is able to move his/her body from one point to the other
Balance	-	the ability to maintain the equilibrium of the body
Strength	-	the maximal one foot effort that can be exerted against resistance
Flexibility	-	the functional capacity of the joints to move through a full range of movement

Assessment

- Identify ways of keeping the body in good health and fit.
- Give advantages of being healthy and fit.
- Explain ways of exercising.
- Give reasons why exercise can contribute to good health.

MODULE 2

COMMUNICATION

UNIT 2.1 DRAWING

Shapes, Forms and Textures

Learners should be able to:

identify types of lines as vertical, horizontal, diagonal, zigzag and curved.

use lines to create contour drawings.

describe form.

explore the effects of light on objects.

explore various techniques of creating tonal values.

describe textures of various objects.

create textures using various drawing tools.

create a composition using tonal values to suggest form.

Shapes, forms and textures

Types of lines

These are vertical, horizontal, diagonal, zigzag and curved lines. Lines could be thin, thick, thin to thick and light to dark. Lines could be found in both the human-made and natural environment

Contour drawings

These are drawings that only show the outline of a shape, or a line that represents the edge of a form. Studies of objects could be done and the main shapes or structural lines could be drawn using different tools and materials. The details such as textural and other surface qualities such as lines and patterns are not necessary.

Shape

Shape is an area enclosed by line. Shapes could be classified as either geometric or freeform.

Form

Form is a three dimensional shape. It has height, width and depths. It also refers to the illusion of three-dimensional shape. A flat artwork as in drawing and painting could be made and could be shaded in such a way that it looks like it has form.

Light and dark effects

The lightness and darkness in drawings of objects can give the illusion of three-dimension on a flat surface. The lightness and darkness could be created using lines and or dots. These could be grouped to show darkness or scattered to show lightness.

Effects of light on objects

Three-dimensional objects have both light areas and dark areas. The light area is where the light falls and the dark areas is where there is no light.

Tonal values
These are the lightness and darkness of black or any colour in art artwork. Tonal values are sometimes used when drawing or painting to give the objects form.

Textures
Texture is the look and the feel of the surface or the surface quality of an object. The surface could be rough, smooth, soft, and hard.

Compositions
The pictures are created either from memory, observation or imagination. Lines, shapes and colours are arranged to create a design or a picture. The themes for the composition could be about a village scene, city/townscape, cattle post, kgotla meeting, independence celebrations, ceremonies, any other life experiences and still-life.

Composition from imagination
This is the act of drawing new images of what has never being experienced before. It could also be the combination previous experiences with new ideas. These are pictured in the mind e.g. floods in desert animals, animal in a war and then drawn.

Composition from observation
It is when looking carefully at an object or a scene and record exactly what you see, such as the surface quality and the appearance of the object that is; its shapes, textures and lines. The drawing could be of a scene, an object or group of objects such as a market place, sports competition, school grounds and buildings or an arrangement of kitchen utensils or tools.

Composition from memory
This is when one recalls the events seen or experienced in the past such as people dancing along the road or in a disco hall, a police station, village shops, kgotla meeting, and lands. These are pictured back in the mind and then drawn either on paper, cardboard or cloth.

Theme
This refers to a topic/idea such as (at the cattle post, disco, lands, independence day, ball game competitions) for a drawing composition. Composition could be either from observation, memory or imagination.

Suggested Teaching and Learning Activities
- Look around your environment and draw objects that have interesting lines e.g. trees and tree bark and grass.
- Observe and record the main lines and shapes of different objects.
- Create contour drawings.
- Observe and draw a contour of an object either natural or man-made and use a range of tonal values to create the form of an object.
- Create a contour drawing using the etching technique.

Support Materials
⇒ Rubbers, papers, erasers, sticks, brushes, cloth and money clips.

Glossary

Shape	-	an area enclosed by line. There are freeform and geometric shapes.
Composition	-	an arrangement of parts in a work of art. These are lines, shapes, and colours and are usually arranged according to the art principles such as pattern, balance and emphasis.

Assessment

- Explain form, shape and texture. Shown in a work of art.
- Describe how form has been created.
- Analyse a given work of art and describe how shapes and textures have been used.

UNIT 2.2: PAINTING

Colour

Learners should be able to:

define colour, value and intensity.

classify colours into primary, secondary and complimentary.

explore various colours to determine values and intensities.

create an arrangement of colours on the colour wheel.

analyse the use of colour in a composition.

Painting techniques

Learners should be able to:

explore the use of various materials, tools and media in painting.

explore various techniques in painting.

create a composition using various painting techniques.

analyse the use of various techniques in a composition.

Colour

Colour is the way something looks when light strikes it. Objects in the man-made and natural environment such as cars, buildings, plants and clothes have different beautiful colours. Colour could be classified into different categories such as primary, secondary and complementary colours. Complementary colours are colours that are opposite each other on the colour wheel.

Value

These are the lightness and darkness of black or any colour in art artwork. Tonal values are sometimes used when drawing or painting to give the objects form. The value could range from the black to white or from the darkest shade of a given colour to its brightest shade.(for example the value of blue ranges from very dark to very light. The lighter values are called tints and the darker values are called shades. Black is added to colour to create a dark shade and white added to colour to it lighter

Intensity

Intensity is the lightness or brightness of colour. A pure colour is called a high intensity and dull colour is called low intensity colour. Once a pure colour is mixed with another

colour, the original colour will become less intense, for example when is yellow added to violet, it will it violet less intense.

Painting techniques

The different painting techniques could be used to create different effects. These would help in expressing ideas, feelings thoughts and experiences. Below is a sample of some of the painting techniques that could be explored:

Creating textures by repeating dots or the use of a stiff brush could also create texture.
Sponging: A wet cloth, brush or sponge could be used to lift paint from the surface to create varied forms.
The use of a dry brush on dry paper will create bold marks.
The application of wet paint onto wet paper will create blending shapes.
Experiment with the varied techniques discussed above on different surfaces to discover different effects. The surface could be such as; paper, bark, glass, cloth, sack and wall.

Composition

Compositions from memory, observation and imagination have been discussed under drawing. These follow the same pattern as that of drawing except that co lour would now be used to create pictures.

In the analysis of different compositions elements such as lines, shapes, colours and textures could be identified as well as painting techniques.

Imagination

This would be the interpretation of ideas/thoughts/feelings/stories/song as well as dreams and create pictorial composition. These are expressed through the use of colour.

Memory

Involves mental recall of experiences or past events e.g. a football match, wedding ceremony (everyday life activities). In painting they are expressed through the use of colour.

Observation

Natural or man-made objects are put together to form a composition. The following could also be observed and drawn: a landscape, still life, or daily activities such as people at the market place, at the bus stop. These are expressed through the use of colour.

Suggested Teaching and Learning Activities

- Experiment with different colours and painting techniques.
- Create a colour wheel.
- Make sketches on researches themes/topics.
- Choose and refine a sketch of the theme/topic selected.
- Create a composition using different techniques and colours.
- Discuss and analyse a given composition.

Support Materials

⇒ Paints, brushes, pallets, pallet knifes, paper, boards, sticks, feathers, leaves, barks, toothbrushes, canvas and visual aids of compositions from memory, imagination and observation.

Glossary

Technique - working skillfully with materials

Experiment - to explore the possibilities and alternatives of media and topic/theme this would be an idea that one has in mind.

Assessment

- Describe the use of colour in a given work of art.
- Analyse a given painting.

UNIT 2.3 TWO-DIMENSIONAL CRAFTS

Collage, mosaic and montage

Learners should be able to:

describe mosaic, collage and montage.

identify materials used for mosaic, collage and montage.

create pictorial composition using either mosaic, collage or montage technique.

Collage

A design or picture made by pasting different cut off materials on a surface. The surface could be paper, cardboard, wooden boards and cloth. The materials are arranged and pasted to create a well-planned design or picture. These materials could either be natural or man-made.

The ideas/thoughts/feelings/stories/songs as well as dreams are interpreted and expressed through a pictorial composition. The themes could also be such as: independence day, cultural day, at the cattle post and at the market. These are expressed through the use of colour.

Collage materials

Both man-made and natural materials could be used to make a composition. The man-made materials could be papers such as; sweet-wrappers, tissue papers, newspapers, cardboard, magazine papers, and other materials such as wool, cloth and threads. The natural materials would be wood, tree barks, grass and twigs.

Mosaic

Mosaic is a design made up of materials such beads, small stones, sand, crushed eggshells. These could be embedded in muddy surfaces such as cement, or glued to surfaces such as paper, boards and walls. Other containers could also be decorated using the mosaic technique.

The mosaic techniques can also be used to decorate walls, ceilings, pavements as well as pictures for hanging. Other works of art such as clay-pots, bottles could be decorated with mosaic.

Mosaic materials

In mosaic making, one type of material can be used and varied in sizes and colours. The small stones or shells can be painted and varnished and used to create a mosaic composition. Materials such as grains, beans and seeds could also be used. Newspapers,

magazines and coloured papers are cut into small sizes and used to create very interesting mosaics composition.

Tools for mosaic making
These would be such as paintbrushes, toothbrushes, knives, scissors and paint-palette (bowls for mixing paints).

Montage

In montage making pictures from magazines, photographs newspapers are cut and arranged on a prepared surface to create a composition. Compositions from memory or imagination could be created by sticking together cutouts pictures in an interesting way.

Materials for montage
Materials such as newspapers, magazines and photographs could be used.

Tools for montage making
These would be such as paintbrushes, toothbrushes, knives, scissors and paint-palette (bowls for mixing paints).

Suggested Teaching and Learning Activities

- Experiment with the collage/mosaic and montage making techniques.
- Make sketches on researches themes/topics for your collage/mosaic and montage making.
- Choose and refine a sketch of the theme/topic selected.
- Create a composition using the different techniques of collage/mosaic and or montage making.

Support Materials

⇒ Wood, cardboard, papers, magazines pictures, cloth, seeds, glue, collected found objects, wool and cotton, sand, eggshells, twigs, grain and beans.

Glossary

Collage - is a picture making by pasting various materials on a surface.
Mosaic - is a decorative design made up of materials such as coloured glass, beads, tiny eggshells, stones embedded in cement, plaster or glued to a surface.
Montage - a picture of art composed by assembling, overlaying, and overlapping many different materials or pieces collected from different sources for example, photographs, magazines and other pictures.

Assessment

- Compare and contrast mosaic/ collage and montage.
- Analyse a given collage/mosaic or montage composition.

UNIT 2.4 MEDIA

Visual communication

Learners should be able to:

explain visual forms of communication.

discuss the advantages and disadvantages of visual communication.

explain the use of visual communication.

design any form of visual communication to convey a specific message.

The teacher should discuss passing information by word of mouth (orally) and by writing. Examples of both written and oral communication can be used to help learners to grasp the meaning of communication. Forms of visual communication – posters, films and videos and charts and diagrams – should be introduced as other ways of passing on a message or information.

The advantages can be its attractiveness, direct appeal especially if bright colours are used, and can be used to explain written messages, especially charts and diagrams. A person who cannot read or write can understand some forms of visual communication.

The disadvantages are that not all people can easily understand charts and diagrams. People may also concentrate on the attractiveness of the film or video but not on the message. Moreover, in order to make sure the receiver understood the message, especially from a film or video, other forms of communication have to be used.

Visual communication can be used to inform people about a new product, hazardous places, dangerous equipment/machines, etc. Posters, for example, can be used to explain messages. It can also show the danger of leaving lights switched on. The relationships in a large data can be simplified in a chart or diagram.

Suggested Teaching and Learning Activities

☞ Ask learners to bring pictures of people transferring messages or information. The emphasis should be sending messages or spreading information. Discuss the pictures with them. Show one group of learners a video carrying a message. Convey the same message to another group orally. Let each group tell the message to the class.

- Show learners some figures and ask them to say what the figures tell them.
- Show learners the graph and diagrams of the same figures and ask them to describe what the graphs and diagrams tell them.
- Describe how a message can be transformed into a poster.
- Let learners identify areas in the school where they can label by posters, e.g. male and female toilets, dining area, dangerous switches, etc.
- Let learners design a poster to advertise School Sport Day.
- Let learners design a sweatshirt logo.

Support Materials

⇒ VCR and cassettes (pre-recorded), newspapers, charts and diagrams

Glossary

Diagram	-	a simple plan to show how something works.
Graph	-	a picture showing relationship between two sets of information. Graph = chart.
Visual communication	-	messages received through sight.

Assessment

- Give two advantages of visual communication.
- Give one disadvantage of visual communication.
- Design a poster to send a message.

MODULE 3

COMPOSING AND PERFORMING

UNIT 3.1 TRACK AND FIELD EVENTS

Sprints

Learners should be able to:

identify sprint events as 100 m, 200 m and 300 m.

recognise starting and finishing points for the selected sprints.

demonstrate the bunch start and running through the tape finish.

apply rules and techniques in any selected sprints.

Sprints refer to short races run at very high speed. The short races ranges from 60m to 400m. These events provide learners with the opportunity to explore and gauge their potentials. The teacher should focus on the following races: 100m, 200m and 300m. Note that the rules should apply to starting points for each race, how to start the race, run and finish a race.

Safety measures should be in place at all times.

Suggested Teaching and Learning Activities

- Practice individually or as groups the following: 100m, 200m and 300m.
- Practice starts and finishes for the selected races.
- Apply rules for the selected races.

Support Materials

⇒ Whistle, lime and rope

Assessment

- Demonstrate how to run 100m, 200m, and 300m.
- Demonstrate starts and finishes for the selected races.
- Apply rules for the selected races.

Throws

Learners should be able to:

identify throwing events as discus, shot put and javelin.

demonstrate proper execution of the selected throw.

apply rules and techniques in the selected throw.

recognise the throwing sector for the selected throw.

Throws are field events that include shot put, javelin and discus. The throws have different throwing areas and these should be used even if modified. The teacher should also practice with learners techniques such as standing throw and turn for discus and shot put. The ways of griping the shot, discus and javelin should also be demonstrated,

Safety measures should be in place at all times.

Suggested Teaching and Learning Activities

- Practice the following ways of throwing:
  - Demonstrate throwing a ball from 2m, 4m and 6m using the overhead throw.
  - Throw a ball from 2m, 4m and 6m using the side arm throw.
  - Throw long enough sticks at varied distances.

Support Materials

⇒ Shot put, javelin, discus and tennis balls

Assessment

- Throw any given implement at given distances.
- Measure individual and peer throws.
- Demonstrate basic techniques for each throw.
- Identify different types of throws.

Jumps

Learners should be able to:

identify jumping events as high, long and triple jump.

recognise the jumping areas for the selected jump.

demonstrate take off and landing in the selected jump.

demonstrate techniques of jumping for the selected jump.

apply rules and techniques for the selected jump.

Jumps are examples of field events in athletics. These include high jump, long and triple jump. Each jump has techniques or styles that are used. For example:

- Long jump - sail techniques
- High jump - scissor/western roll
- Trip jump - hang

It is important to exercise a lot of safety in jumps to prevent injuries.

Suggested Teaching and Learning Activities

Practice the following ways of jumping:

- Jump from both feet
- Jump from one foot
- Jump in the air and taking off with both feet Perform a standing long jump
- Jump from a high object such as chair Use a rope to jump off with one foot and both feet
- Jump with one foot following the other (like a scissor) on the ground
- Jump high and low changing shapes
- Demonstrate throwing a ball from 2m, 4m and 6m using the over head throw
- Throw a ball from 2m, 4m and 6m using the side arm throw

Practice the following ways of jumping and landing:

Jump on 1 foot or both

- Jump with arms raised
- Jump with 1 and 2 feet from a small brick
- Jump high and low changing shapes

Assessment

- Demonstrate a jump from both feet
- Demonstrate a jump from one foot
- Demonstrate a jump in the air and taking off with both feet
- Perform a standing long jump

UNIT 3.2 CREATIVE DANCE

Dance elements

Learners should be able to:

identify dance elements as level, shape, direction and pathways.

discuss aspects of body movements as body awareness and space awareness.

demonstrate the selected dance elements.

demonstrate aspects of body and space awareness.

explain choreography.

demonstrate proper use of personal and general space.

show co-operative and social skills in dance.

use selected dance elements to create a short dance.

show formations and patterns in dance.

explain the role of dance in the community.

evaluate compositions using a set criteria.

Elements of dance [What makes up a dance]

There are many elements that make up a dance. Below are examples of elements that can be used at standard three. The teacher has to explain each element and demonstrate it to the learners.

Levels	-	low position e.g. squatting
	-	medium position e.g. leaning/bending/standing with knees bent
	-	high position e.g. upright position with arms raised or dropped
Shapes	-	angular, circular, round, curved, long and short
Directions	-	Forward, backward and sideways movements
Pathways	-	zigzag, curved and diagonal movements

Use of space

It is important that learner are shown how to use their own space, that is where they are able to move freely without touching or space within reach as well as the general space that which they share with others or objects. This will reduce injuries and will teach learners the importance of utilising efficiently whatever space is available to them. When learners use any kind of available space, they should take into account, aspects of body movements as well as dance elements in response to the different stimuli. Learners should be encouraged to come up with their own short simple dances to show that they can be creative.

Suggested Teaching and Learning Activities

- Move in general and personal space alone and with others.
- Use personal and general space efficiently.
- Use different parts of the body to move in space such as:

  - walking - on heels, toes, knees bent, knees stiff, legs kicking up high in front back and side
  - arms - held at the side, overhead, folded at the back/front, swinging, straight etc.
  - body position - bent forward, backwards or side ways
  - changes in position - in one place, forward, backwards/side ways
  - levels - on tip toes, knees, half bent etc.
  - size of steps - short or long
  - timing - fast or slow movements
  - changes in force - heavy, light or sustained movements/steps
  - movement quality - smooth, flowing or sharp and jerky steps
  - other movements - running, jumping leaping, hopping, galloping, sliding, skipping and rolling.

- Take into account body, effort and space awareness and relationships when moving and establish rhythm.

Body movements

- Swing the body use one arm, two arms, one leg or trunk
- Rock side ways changing weight from one side to the other front /back
- Push with one hand, two hands one foot and two feet, hips and shoulders
- Pull with one or both hands, one or both feet, forward, backward, sideways or downwards
- Strike with one or both arms and alternate them
- Lift with one hand or both

Body control, balance, shape, pathways, directions and coordination

- Soft movements - Balloons, birds flying, clouds and leaves falling.
- Heavy strong movements - galloping, chopping wood and digging.
- Sharp jerky movements - frogs, grasshopper moving, machine and robot.
- Upwards and down wards movements- bouncing balls, opening an umbrella, stamping, sediba sa terai (petse), see-saw and pumping.
- Movements high – aeroplane and climbing a tree.
- Fast movements - motorcars, dogs running and galloping horses.
- Slow movements - pushing, pulling a heavy object, paddling a heavy canoe and pushing a heavy object.

Support Materials

⇒ A variety of recorded music (loud to soft), guitar, whistle, drums, rattles, stones, (matlhao), cans, beads, videos tapes of traditional dances, radio cassette recorder, television set and video recorder.

Glossary

Being still/stillness	-	to be freeze position or not moving.
Sustained movements	-	controlled, positive, continuous or constant.
Levels	-	different body positions in space.
Locality	-	an area/place where one lives.
Unison	-	moving together in a group, at the same time.
Canon	-	a way of overlapping movements at intervals by more than one dancer or following others as they move.
Space	-	the element one occupies and moves through.
Choreography	-	choosing, arranging and combining movements together to create a dance form.

Assessment

- Identify and perform elements of dance.
- Demonstrate proper use of personal and general space.
- Use themes to create a dance piece.
- Identify and travel on different parts of the body.
- Demonstrate different aspects of body movements.
- Demonstrate a variety of patterns as you dance in small groups.
- Make patterns and formations as they dance.

Traditional dance

Learners should be able to:

3.2.3.1 name traditional dances in Botswana.

3.2.3.2 list communities associated with each dance style.

3.2.3.3 perform traditional dance styles in a given locality/district.

3.2.3.4 explain the significance of each dance.

3.2.3.5 differentiate between dance patterns.

Learners should be allowed to perform traditional dances within their locality and appreciate them. There are many different traditional dances for men and women. These will vary from locality to locality/district to district. It is therefore suggested that learners dance and sing traditional dances known to them or common to where they live. Learners have to include the elements of dance in their own dances to come up with a new dance piece. This can be based on themes such as:

◆ Life at the lands	-	ploughing, weeding, chopping trees, harvesting, reaping,
◆		stamping, etc.
◆ Animal imitation	-	goats, cows, donkeys, horses, giraffes, lions, crocodiles, hippos etc.
◆ River flowing	-	swiftly or roughly
◆ Rain clouds gathers	-	coming from different directions, in different shapes and colours

Different types of stimuli can be used so that learners respond to different sounds. Learners can also be introduced to traditional dance songs such as musical games like koi. Different types of skipping (koi) can be played in singles and doubles with accompaniments (singing).

Examples of koi [Skipping as a game with singing]
Petoro, maloba, pena, be iwee, seteke, nna le rammata, re tswa go sila mealie-meal etc.

Suggested Teaching and Learning Activities

Use a drum/cassette tape to suggest the mood and rhythm for any movement

- Perform traveling actions responding to the stimuli of choice.
- Vary directions, levels, shape and speed when moving.
- Perform singing dances in their locality or district e.g. koi, terena, Ngwedi le letsatsi, bana tlang gae etc.
- Practice and perform movement patterns of selected traditional dances such as tsutsube, setapa, borankana, ndazola, phathisi etc.
- Perform musical games such as terena tshukutshuku, maburu wee dichankana, mmupudu o wa wa, a nka go kitla ka letswele lame, I want to see my Jane etc.

Support Materials

⇒ A variety of recorded music (loud to soft) - guitar, whistle, drums, rattles, stones, (matlhao), cans, beads, videos tapes of traditional dances, radio cassette recorder, television set and video recorder.

Assessment

- Identify and Perform traditional dances in their locality and district.
- Identify and perform aspects of the body movements.
- Identify and perform elements of dance.
- Demonstrate proper use of personal and general space.
- Use themes to create a dance piece.
- Identify and travel on different parts of the body.
- Demonstrate different aspects of body movements.
- Demonstrate a variety of patterns as you dance in small groups.
- Make patterns and formations as they dance.

UNIT 3.3 GYMNASTICS

Gymnastic movements

Learners should be able to:

identify gymnastic movements.
perform gymnastic movements of travelling, balancing and landing.
explain linking movements as leaps, turns and jumps.
use linking movements of leaps, turns and jumps in a sequence.
perform counter balance and counter tension skills with variations.
perform rhythmic movements in gymnastics.
perform a short sequence alone or with a partner in unison and canon.
observe and copy a partner.
evaluate performance by describing.

Gymnastics refer to a wide variety of physical activities in which learners develop confidence, control and coordination skills to be able to manage their bodies in different situations. It provides learners with the opportunity to explore the capabilities of their

bodies, promotes creative and inventive experiences. There are two forms of gymnastics and these are rhythmic and artistic.

Basic gymnastic movements are important for beginners to start with in order to aid movement development and prevent injuries. Learners can be introduced to gymnastic basic aspects of body through movement. Movement exploration should be encouraged without restrictions in order to achieve creative and inventive experiences and ideas.

Learners can be introduced to traveling on different parts of the body as well as taking weight on different parts of the body. Basic balancing skills can also be re-introduced at standard three, such as static, inverted, counter balances, counter tension and matching balances.

Safety measures should be in place at all times. Learners who are not able to perform certain skills should be spotted properly to aid movement and prevent injuries. As part of safety learners should be taught how to perform break falling or break a fall from a balance.

Suggested Teaching and Learning Activities

Travelling Activities: Alone and in pairs, walk on different parts of the body as in dance at different levels e.g.

- Travel with your partner touching and not – walking, running, jumping, skipping etc.
- Run from point A – B and be still when required.
- Join hands with your partner and travel together with a partner perform a wheelbarrow.
- Balance while moving with a beanbag on the head.
- Perform forward and backward rolls in a straight line.
- Move with weight on others e.g. carry each other on your backs i.e. Back to back like tea/coffee game.
- Move with a partner slowly/quickly/rhymically matching movements.

Jumping Activities

- Jump with or without ropes.
- Follow your partner copying her/him.
- Spring and land as a frog over and object.
- Roll forward, lift an object and roll.
- Tumble through a hoop.
- Perform jumps such as straddle, jump, arch jump, cat leap.
- Run and change directions and speed alone and with others before a jump.

Balancing Activities

- Place a beanbag on the head- go to the market.
- Run with a beanbag on the head.
- Move about space in small and large imitating animals such as a cat, a donkey and a lion.
- Inverted balances e.g. hands and head stand, cartwheel.
- Counter balances – balancing on different parts with a partner.
- Make partner balances of your choice.
- Horizontal balances – 2 to 3 points on contact with the floor, etc.
- Partner rock on different directions.
- Partner frog leap in unison / canon.

Jumping and Landing

- Jump on one foot or both.
- Jump with arms raised.
- Jump with one and two feet from a small brick.
- Jump on one foot and land on two feet.
- Jump low and turn.
- Straddle, pike, straight and star jump.
- Cat leap, quarter and half leap.
- Partner jump over a rope unison/canon.

Support Materials

⇒ Gymnastic Floor mats, 10 stock bricks, benches, hula-hoops, skipping ropes, plastic balls and whistle.

Glossary

Counter tension	-	balancing of forces/weights in opposition e.g. tea/coffee game.
Static balances	-	is when the body weight is supported by lower body parts; upper arms or head such as feet/knee or bottoms with the shoulders above the point of support.
Inverted balances	-	when the body weight is supported by either the hands, upper arms or head with shoulders above the point of support.
Counter balances	-	weight used to balance another weight or offsets another e.g. as pairs pushing upwards from a low position in the same direction at the same time.
Rhythmic gymnastics	-	flow of movement characterised by different features or elements in alternation. These have repetitions, timing, spacing, etc.
Artistic gymnastics	-	gymnastic skills performed with skill and appreciation of beauty from within.

Assessment

- Perform different ways of moving.
- Perform individual and partner balances.
- Perform different jumps with variations and shapes.
- Match balances.
- Perform inverted, vertical and horizontal, counter balance counter tension, matching balances.
- Jump and land on one and both feet.
- Identify games which have chasing and running with varied speed and pathways.
- Jump low and make turn half, quarter and full turns.
- Straddle jump.
- Cat-leap and jump with a partner in unison.
- Partner jump over the rope in unison/canon.
- Travel with your partner touching and not touching as in walking, running, jumping, skipping etc.
- Run from point A – B and be still when required.
- Join hands with your partner and travel together with a partner perform a wheelbarrow.
- Balance while moving with a beanbag on the head.
- Perform forward and backward rolls in a straight line.
- Perform a cartwheel.

- Move with weight on others e.g. carry each other on your backs i.e. Back to back like tea/coffee game.
- Move with a partner slowly/quickly/rhymically matching movements.

UNIT 3.4 BALL GAMES

Invasion/contact games

Learners should be able to:

identify invasion/contact games as netball, football and basketball.
explain sending and receiving skills for the selected game.
perform sending and receiving skills for the selected game.
recognise court dimensions and positions for the selected game.
identify pieces of equipment used for playing the selected game.
integrate skills in the selected invasion/contact games.

3.4.1.7 *demonstrate offensive and defensive skills.*

3.4.1.8 *apply rules in a modified selected game.*
play an even-sided modified game.

explain physical, social and mental benefits of participating in games.
demonstrate interpersonal and social skills in playing games.
utilise available time to participate in physical and social games.
demonstrate co-operative and decision making skills in games.
list recreational and social values of games.
explain the importance of good teamwork spirit.
demonstrate good teamwork spirit in games.

There are different games that learners can participate in. Examples are sport skills, lead up, co operative and recreational games. Learners should be encouraged to practice, perform and refine skills in any of the selected games. Lead up games, cooperative and recreational games can be played to enhance sport lead games. These games encompass locomotor, manipulative and stability skills and are characterised by sending and receiving of the ball/implement. Selected games can be played to develop sending and receiving skills such as throwing and catching, striking, batting, fielding, kicking and trapping.

Examples of sending and receiving skills include catching, throwing, drifting, pat bouncing, passing, kicking shooting, striking and rolling.

Examples of traveling movements include walking, running, jumping, rolling, skipping, turning, sliding, leaping and galloping etc.

As in dance, learners' attention should be brought to aspects of body movements and movement patterns.

Experiences and skills

Explore freely and practice sending and receiving skills of throwing and catching (bouncing, rolling, fielding hitting a target). The following throws should be practiced

- underarm which is similar to rolling or tossing
- over-arm which is similar to throwing an implement from above the head onto the ground

Exploration and free practice with right and left hand and both
- at different levels of the body
- to either side of the body
- in different body positions, with a step or jump
- catching while moving
- add a variety of movements such turning ,clapping , jumping before catching a ball

Application

Individually or with a partner, catch a ball that has been:
- dropped by one person.
- thrown by one person.
- thrown by a partner.
- thrown around in a circle.

Suggested Teaching and Learning Activities

Activity 1

Throwing and catching through a hoop, A and B throws the ball and then C holds the hoop. Count the number of successful throws one can make out of 10 throws and change roses.

Activity 2

In twos, one child holds the hoop while standing on the chair/bench or brick. The other tries to throw the ball so that it can drop through the hoop, change responsibilities after 5 balls.

Activity 3

Catching a bird - In two or threes A, B and C .A tries to throw to B, while C intercepts. Those with the ball or beanbag, can change directions and positions and C will move accordingly.

Activity 4

Dribbling - These are short kicks forward which keeps the ball close to the player's foot all the time. Dribble alone and with a partner in a straight line and between objects. Dribble in pairs and around an object. Increase the distances between as you dribble.

Activity 5

Kicking –Concentration should be on using the instep kick and inside and outside parts of the foot. Pass the ball to a partner slowly and varied speed while stationary and moving score between the partners legs or cones. Increase the distance and kick to the partner accurately or design passing relays/double relays in small groups.

Activity 6

Trapping – alone and in pairs use a dribble relay to pass and stop the ball. Stop the ball with the right, left foot and both.

- Perform traveling activities with different parts of the body.
- Change speed shape and levels when moving.
- Use general and personal space properly and efficiently.
- Play or perform recreational, lead up, co-operative and sport skill games.
- Throw a beanbag or ball at a distance/target with accuracy and speed.
- Use the under arm and over hand to throw a ball/bean bag or implement.
- Throw a ball/bean bag or implement for height.

- Throw to a partner at a stationery position and when moving.
- Play a throwing and catching as well kicking and trapping game.
- Strike a ball or beanbag with hand bat or racquet.
- Bounce and the ball into a free space.
- Pat-bounce the ball on one spot and change hands while moving.
- Bounce the ball against the wall and catch.
- Bounce the ball to a rhythm and hit it with the palm.
- Bounce the ball and kick it.
- Stop a kicked/struck ball.

Support Materials

⇒ Beanbags, plastic balls, tennis balls, home made balls, hula-hoops, Frisbee, bibbs, skipping ropes, whistle, drum and cones.

Glossary

Locomotor	-	ways of moving/travelling from one place to another.
Manipulative	-	work/operate something with the use of hands or feet.
Stability	-	quality of maintaining firmness or force on different body parts and ways.
Sport lead games	-	a combination of fundamental movement, patterns or skills applied to perform a sport related activity.
Lead up games	-	minor games leading to acquisition of basic sport skills.
Co-operative games	-	these are games played for enjoyment and or fun, non-competitive.
Recreational games	-	form of play used as game for relaxation or amusement.

Assessment

- Identify selected sending and receiving skills.
- Perform sending and receiving skills alone and with a partner.
- State basic rules for selected games.
- Stop /catch a ball.
- Alternate hands feet when bouncing and kicking the ball.
- Demonstrate the over hand and under arm throws.

UNIT 3.5 MUSIC OF BOTSWANA

Composers and artists

Learners should be able to:

differentiate between a composer and an artist.

list famous composers of Botswana origin.

classify famous compositions according to composers.

name famous traditional artists of Botswana origin.

list famous artists of modern music in Botswana.

perform any composition by a famous composer/artist.

discuss the significance of music in society.

A composer is someone who writes music but an artist is a professional performer of music dance and drama. Composing deals with originality, creativity and imagination of the product. Artists on the other hand deals only with creativity and imagination, their products

are therefore not originally theirs. Artists can in some cases become composers if they manipulates and arrange the media unique to a particular art to come up with an expressive original product.

Famous composers	Famous artists of modern music
K.T. Motsete Duncan Senyatos B. L. Bogosing Ricks Morake G. Motswaledi Soccer Moruakgomo Frank Lesokwane Alfred Mosimanegape Ashley Gops Luka Disho Mmereki Marakakgoro R.N. Ragalase Bafana Pheto Molefe Molefe	Vee Maxy Mega V Duncan Senyatso Soccer Moruakgomo Ndingo Jowa Ashley Gops Machesa Kgobola Alfredo Mos Sca Oracle Kast Lily Mr. Tagg Stagga Skizo Eskimos Take five Fanco Tu Unik Unik Attractions Wizards of the desert Chashless society Donald Botshelo Devine T-Joint
Traditional artists	
Speech Madimabe Ratsie Setlhako Bachoni Matsoni Teemane Matlapeng Setampore Kotaeshele	

Significance of music in society

Music plays a very important role in the live of man. It is an essential part of human life because of the following reasons;

- It entertains
- It communicates
- It can be used as a symbolic representation
- It is a form of emotional Expression
- It stimulates physical response
- It contributes to the stability and changes within cultures

Suggested Teaching and Learning Activities

Activity 1

Ask learners to give names of famous individuals and groups which play music in Botswana. Use the list they have given to show the difference between artists and composers as explained above. From the list ask them to list names of composers and of those who are composers and artists. They can also use the list to identify famous artists of modern music. Let learners cut pictures of these famous musicians and paste them on manila sheets with proper labels.

Activity 2
Divide the learners in groups of three and ask each group to choose any famous local song. After choosing the song they have to practice and perform the song or using the song to the whole class. Make sure they do not all choose one song to ensure variety. During their performances let them state the name of the song, the person singing it and how their performance is going to be. To safe time limit the performances to only 2 minutes.

Activity 3
Let learners role-play situations where music is played or sung. Discuss the role played by music in each situation. Try to fit the role in the categories discussed above under content. Divide them in groups and give them different tunes of music to role-play. Let each group present to the class.

Support Materials
⇒ Magazines, newspapers, radio tape or CD player, tapes and CDs

Glossary
Composer - someone who writes music or someone who performs original music from his creativity.
Artist - a professional performer of music.

Assessment
- State the difference between a composer and an artist.
- List famous composers of Botswana origin
- Classify famous compositions according to composers
- Name famous traditional artists of Botswana
- List famous artists of modern music in Botswana
- State the importance of music in our society

UNIT 3.6 PITCH

Pentatonic scale
Learners should be able to:

describe pentatonic scale.

sing notes of the pentatonic scale in correct pitch.

play notes of the pentatonic scale on pitched percussions.

perform melodies in pentatonic scale.

A pentatonic scale is a scale of five notes widely found in folk music. It can be easily produced by playing the five black keys on a keyboard starting with the group of three ending with the group of two. This scale avoids the use of the half steps between m-f and t-d'. It then gives the sound of d r m s l. These five note forms the pentatonic scale. It sounds almost like a tune and any order of the notes has its own beauty.

An example of a song in pentatonic scale is 'Nobody knows the trouble I have seen'. This scale is believed to be the native scale of Africans.

d r m s l d'

Pitched percussions are instruments like marimbas, xylophones, metallophones and glockenspiel. They are percussion instruments with defined pitch.

Suggested Teaching and Learning Activities

With the understanding of the above content all the specific objectives will act as activities.

UNIT 3.7 NOTATION

Types of notation

Learners should be able to:

explain notation.

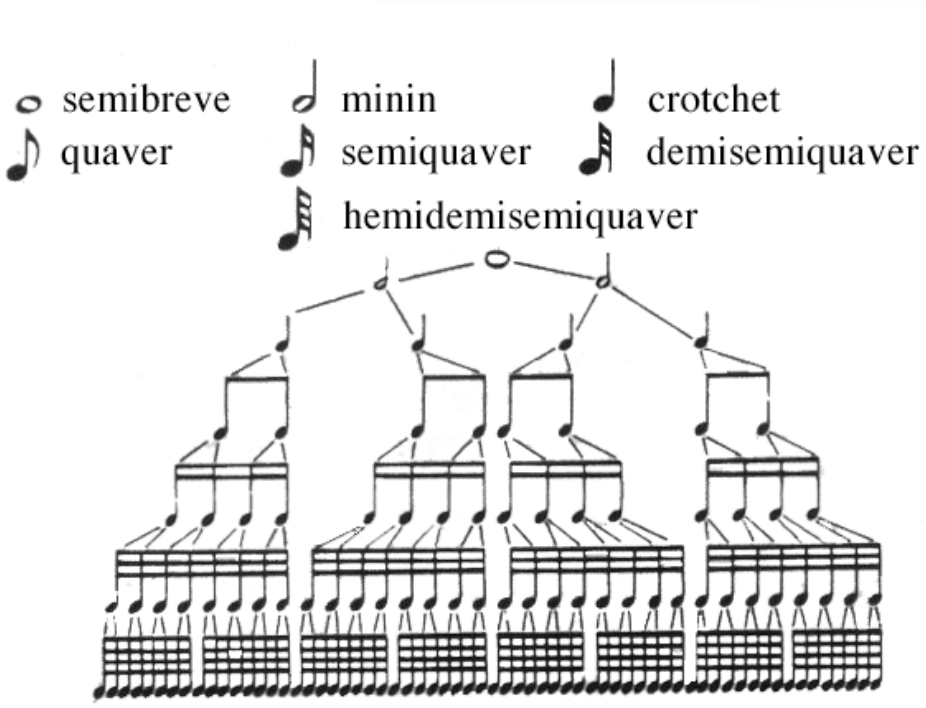
describe types of notation.

create graphic notation of simple tunes.

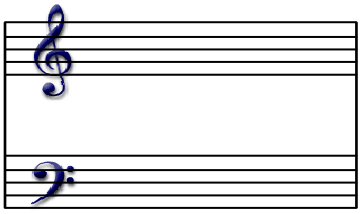
state the importance of notation.

Notation is a system of written marks or signs used to represent music. This system is used to write down music so that it can be performed. These systems of notation are purely European in origin but very useful in Africa. There are three types of music notation namely***, staff notation, tonic solfa and graphic notation.***

Staff notation



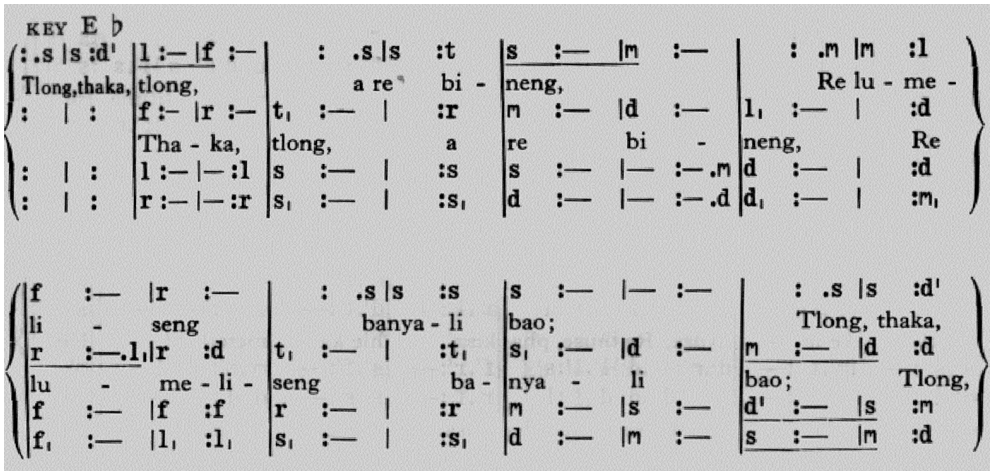
All these signs above are used in staff notation and they help identify the duration of the sound. This determines the rhythm. The pitch of the music is determined by where these symbols are placed on the five lines and four spaces shown below. These five lines and four spaces are called the stave. The female voices and high-pitched instruments are notated on the upper stave and the male voices and low-pitched instruments are notated on the lower stave.



Tonic solfa notation

This type of notation uses the modulator as its basis for pitch. Different letters are used and their pitch varies. There are basically seven pitches with the first one repeated an octave higher to make a total of eight notes d r m f s l t d'. The rhythm is determined by the use of beats and measures where the notes are placed to determine their duration. Pitch in tonic solfa is relative to the tonic if the tonic changes the pitch of the other notes changes too.

Tlong, Thaka



Sometimes music can be written with both tonic solfa and staff notation on one sheet like in the example bellow.



Bakang, bakang Modimo
(Round ya ditlhopha di le 4)

Bakang, bakang Modimo,
Rra wa legodimo le lefatshe!

Go ka opelwa ka mafoko a mangwe jaaka:

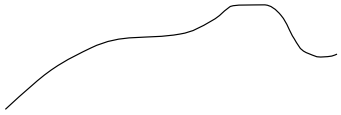
Thobo, bana ba Modimo,
bana ba Modimo, lebogang

Graphic notation

Graphic notation is the representation of pitch, tone duration and dynamics by means of pictures, dots, lines, curves et cetera. This system is especially suitable for class music, as learners do not require pre-knowledge to be able to link these symbols to sound. This was encouraged at lower classes should still be used at this level. It will help pupils understand other notation systems easily.

Line notation is the use of lines to indicate pitch. It can be used with learners who have little or no experience of notation. Line notation as such does not teach learners to read music, but it does make them aware of whether tones are high or low.

A melodic contour by means of line notation



(Kum ba yah, my lord, kum ba yah)

Rhythm patterns by means of long and short lines

– – — - ——— – – ———

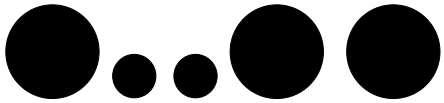
(Kum ba yah, my lord, kum ba yah)

A combination of melodic and rhythmic notation

–
–
— - ——— ———
– –

(Kum ba yah, my lord, kum ba yah)

Make a visual or graphic representation of the name of the character *Betty Martin*.
Learners may repeat this:



Hey - Bet - ty - Mar - tin

UNIT 3.8 TIME AND RHYTHM

Notes

Learners should be able to:

explain rhythm.

identify a crochet and a quaver in staff notation.

draw a crochet and quaver.

state time values of a crochet and quaver.

demonstrate rhythmic combinations of crochet and quaver.

create short rhythmic patterns with a crotchet and a quaver.

perform a full note and a half note in tonic solfa.

draw a full note and a half note in tonic solfa.

create short rhythmic patterns with full and half notes.

Rhythm covers every thing pertaining to the time aspect of the music. It covers the effects of beats, accents, measures, notes and rest values, grouping of notes into beats and grouping of beats into measures.

Notes have three parts which the head; the big round part, the stem; the stick connected to the head and the flag; a small hook which joins the stem and helps notes to be beamed together.



quaver



crotchet



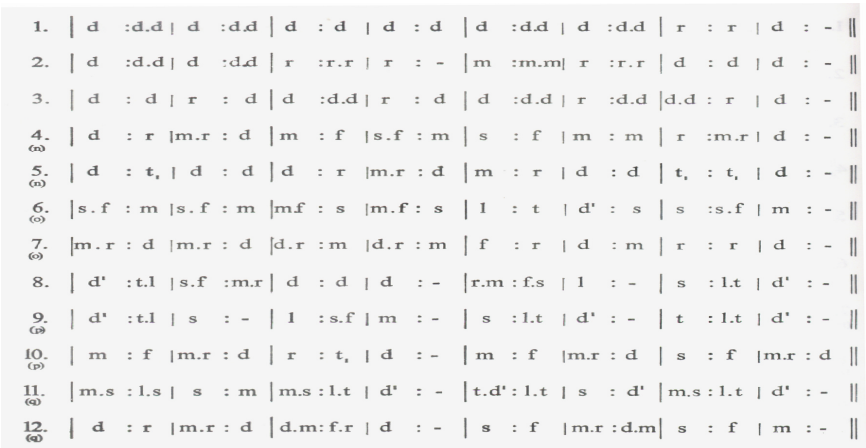
beamed quaver

The note values of these notes depends on the given time signature but for this level the explanation should be given in common time. A crotchet will be equal to one beat and a quaver will be equal to half a beat. This means that two quavers will be equal to one crotchet. The duration of two quavers will be equal to the duration of one crotchet. Two crotchets will be equal to four quavers.

| : this represents a beat in tonic solfa to create a half beat a dot is placed to divide the beat in two halves | . : this shows two half beats.

| d.d : these are half notes within a beat

| d : this is a full note within one beat



These twelve examples shows the use of half notes and full notes in tonic solfa the teacher can write these using crotchets and quavers. These rhythmic combinations can also be clapped by the class. The class can also sing them in tonic solfa. The teacher can therefore use the above examples for many activities of this unit.

UNIT 3.9 DRAMATISATION

Drama

Learners should be able to:

explain drama.

list types of drama.

explain uses of drama.

identify elements of drama.

explain selected elements of drama.

perform selected elements of drama.

appreciate the role of drama in society.

dramatise short stories/tales to communicate an idea.

Drama can be used to express moods, feelings, ideas or thought. When people dramatise stories, they can work on the basis of what exists or imaginary. Drama can be used as a form conveying or communicating a message, an idea and mood of thought. When dramatising it essential to use gestures or movements to express the role of different characters in the story. Traditionally and in modern society, there are stories or tales that can be mimed, dramatised and created from simple tunes such as the story of **the lion king** or **birth of Jesus**. When one is dramatising, there is usually:

- change in tempo of sound (fast and slow)
- change in volume of sound (loud and soft)
- slow and fast movements
- narrative and response movements

Suggested Teaching and Learning Activities

Activity 1

Describe different moods e.g. happy as at a party, wedding and sad at in mourning, lonely, afraid etc. and ask learners to explain how they recognise each of the moods. Ask them to sing a song, play games or make sounds or gestures which depict each mood.

Activity 2

Combine tempo and volume and ask learners to describe moods depicted by the combinations of

- fast and loud
- fast and soft
- slow and soft
- slow and loud

Activity 3

Pick one learner to narrate a well-known story and ask for volunteers to show the mood using gestures or movements to tell the story. The teacher can demonstrate first.

Activity 4

Divide the class into small groups and ask each group to make up a story which they will dramatise with one of them as a narrator. Each group should have a short theme e.g. hunting, harvesting, autumn, parting, mourning at the mines etc.

Support Materials

⇒ Costumes made from plastics, old cloth and cardboards to depict different characters.
⇒ Pictures of different animals and characters of their choice.
⇒ Rattles, drums, hoops, tennis balls and shakers.

Glossary

Mood	-	the way one feels at a particular time.
Narrative	-	something that is told as a story.
Narrator	-	a person who tells a story.
Stimuli	-	something that causes a reaction.
Characters	-	people in a story.

Assessment

- Create different moods a given song.
- Create a short story based on individual theme.
- Respond to cues given by the teacher from a particular story.
- Change moods and personalities of characters in a given story.
- Imitate characters and personalities as read from the given story.

MODULE 4

DESIGN AND PRODUCTION

UNIT 4.1 NEEDS AND WANTS

Individual and household needs and wants

Learners should be able to:

explain needs and wants.

differentiate between needs and wants.

classify school requirements into needs and wants.

discuss ways of contributing towards household needs and wants.

The teacher should discuss the meanings of needs and wants with learners so that they are able to identify human needs with **essentials** but wants with **non-essentials** or ***luxuries.***

Examples of needs

- Food - anything eaten to quench hunger or thirst
- Clothes - anything for covering our nakedness
- Shelter - anything that protects us from the weather – hot and cold temperatures

Examples of wants

- Wants - television, entertainment, car, bicycle, cell phones, etc.

Characteristics of needs and wants

Needs	***Wants***
Required in order to live	Required to make life better
Are few	Are many and keep changing
Usually material	Both material and non-material
Easily satisfied	Difficult to satisfy all

School Needs	***School wants***
Food	Books
Uniform*	Stationery (pens, pencils, etc)
Classroom	Furniture

* Where a school uniform is prescribed, a child without one suffers psychologically even if s/he is not sent home from school.

Learners should be able to explain why each of the above is either a need or a want. Learners should be able to identify family needs. They must be able to tell how they should handle the family television and the music system carefully so they are not damaged.

Suggested Teaching and Learning Activities

- Make a table of needs and wants.
- Let learners name their individual wants.
- List all the wants of individual learners.
- Help learners to compare needs with wants.
- Give the characteristics of needs and let learners compare them with those of wants.

Support Materials

⇒ Products such as food items
⇒ Items of clothing
⇒ Pictures of houses, cars, aeroplanes, bicycles, etc.

Assessment

- Ask learners to list their needs.
- Make a list of wants for school.
- Let learners select needs from a list of needs and wants.
- Identify the providers of their wants.

Satisfaction of needs and wants

Learners should be able to:

explain the satisfaction of needs and wants directly and indirectly.

discuss the part played by money in the satisfaction of needs and wants indirectly.

discuss ways of satisfying household and school needs and wants.

Making and consuming or using goods and services satisfy needs and wants. If we make and consume or use what we have made ourselves, then we have satisfied our needs and wants directly. For example, if we need vegetables and we take vegetables from what has previously been growing in our backyard, we have satisfied our need for vegetables directly. If you need a jersey and you weave the jersey yourself and make use of it, then you have satisfied your need for jersey directly.

In the rural areas, most needs and wants are satisfied directly. A farmer, for example, will use part of the maize he/she has produced, take milk from the cows and slaughter some livestock for meat.

Needs and wants are satisfied indirectly when we exchange some of the things we have for the things of other people. In the village, for example, if one person exchanges meat (which he/she has) for another person's maize, he/she satisfies his/her need for maize indirectly. The exchange of goods for goods is called barter. So the exchange of meat for maize is barter.

Barter exchange is rare these days. What is more common is selling what you have for money, and using the money to buy what you want. A farmer may sell cows to BMC and be paid in money. He then uses the money to buy jersey and other things.

At work, parents may render services for which they are paid money every month. The money is then used to provide the household needs and wants. The local and central governments provide some of the needs and wants for school. For example, books, tables and chairs are provided by the council.

Resources are used to satisfy needs and wants. There are resources from land (trees, grass, water, soil, etc) which can be used in crop growing and livestock farming. Other resources are labour and capital.

Suggested Teaching and Learning Activities

- Ask learners to indicate the providers of their needs and wants.
- Let learners describe how they can contribute to satisfy the family needs or wants.
- Let learners simulate trading in class.
- Let learners explain the part money plays in satisfying needs and wants.
- Discuss those needs and wants of the household that can be satisfied without the use of money.
- Ask learners to complete the table shown below

People's basic needs	Resources for satisfying them

Support Materials

⇒ Money in coins and notes or anything to represent money
⇒ Items of food, clothing, etc.
⇒ Magazines, newspapers, etc.

Glossary

Barter - the exchange of goods for goods
Capital - money, machines and tools used to start a business
Exchange - giving a product or service and receiving money in return.
Labour - the effort of the worker
Resources - a stock or supply of materials.

Assessment

- What is barter?
- Explain indirect satisfaction of wants.
- What part does money play in exchange?

Budgeting

Learners should be able to:

explain budgeting.
explain the importance of budgeting.
analyse the parts of a budget.
discuss effective ways of budgeting.
prepare a family budget.

A budget is a detailed plan of income and expenditure for the future. Budgeting is the preparation of a financial plan for the future. As individuals we need to plan the spending of our future income. Income is the money we receive for working or as gift. In the case of learners, it's the money they receive from parents and other relatives. We need to plan the spending of this income otherwise we may not spend it wisely.

To prepare a budget, we need to be very familiar with all the things we spend our incomes on. We must also know the prices of those things and the quantities we usually buy. We must also be aware of how much incomes we are going to get.

We must then list all the items we want to buy including their quantities and cost. For example, if I hope to get P30.00 to spend from Monday to Friday, I may prepare a budget as follows:

	P	t
Transport to and from school	12	50
2 bottles of fizzy drink	5	00
5 packets of chips	4	50
Payment of club dues	2	00
Payment of a loan	4	00
Savings	2	00
Total	30	00

Budgeting help to:

a) Match expenditure with income.
b) Plan for future resources.
c) Control expenditure.
d) Make future plans.

A budget has two main parts: income and expenditure.

Income deals with the various ways we expect money to come from. If you are a learner, you expect to get money from your parents and other relatives. You may also make some money doing small jobs.

Expenditure shows the things you plan to spend your money on. The income earners in the family may prepare a budget to show how they intend spending the monthly income. The expenditure may be on food, rent, transport, utilities, clothing, school requisites, etc. The aspect of saving must be stressed. Let learners know why the government prepares annual budget.

Suggested Teaching and Learning Activities

- Learners show how to spend P10.00 more effectively.
- Learners list various sources they can get money from in one week.
- Learners list the goods and services they will spend the money on.
- Teacher reads parts of a published government budget to the class.
- Teacher explains expected income and expenditure of the government to the class.
- Learners list sources of income for their homes for one month.
- Learners list the items of expenditure for their homes for one month.
- Learners prepare a family budget for one month.

Support Materials

⇒ Published government annual budget, money and manila paper.

Glossary

Expenditure	-	the spending of money
Financial plan	-	plan for the spending of future incomes
Income	-	money received from working or selling something

Assessment

- Identify ways of making your future spending just equal to your future income.
- List the main items of your expenditure.
- Identify the differences between the main items of expenditure of a family and of a business.
- Explain why some people can spend more than others.
- Explain why saving is important.
- Identify the institutions for keeping savings.

Business

Learners should be able to:

explain business.

analyse different goods and services supplied by businesses.

discuss ways businesses identify the needs and wants of consumers.

examine the part played by business in satisfying needs and wants.

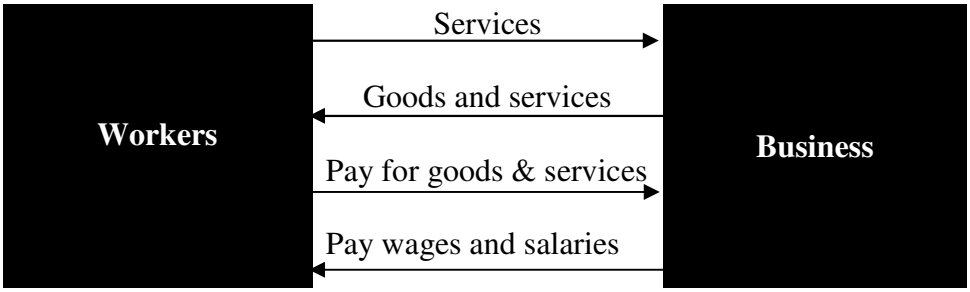
investigate the activities of a specific business in the locality.

Business deals with activities involved in making and distributing goods and rendering of services for profit. Some businesses are engaged in manufacturing, commerce, legal, entertainment, etc. Businesses produce goods and render services. The goods produced can be classified into consumer goods and producer goods.

Similarly services can be commercial or direct personal services. Commercial services involve wholesaling, retailing (buying and selling), banking, insurance, transport, etc. Direct personal services are those of education, health, security, law and order, etc.

Businesses are able to find out the needs and wants of people mainly through market research. Thus, if a business wants to produce a particular product, it will find out from probable consumers whether they want it. This helps the business owner to make the product to suit a particular group of consumers (market).

People satisfy their needs and wants by acquiring and consuming goods and services. The goods and services are provided by businesses. Businesses, in fact, employ people and then pay those wages or salaries for their services. The people employed produce goods and services which are required by other people to satisfy their needs and wants.



The teacher should help the pupils to select some businesses – farming, trading, manufacturing, service – and find out what they do in groups. Few businesses should be selected at this stage.

Suggested Teaching and Learning Activities

- Visit some businesses in the local community to find out what they do.
- Draw a map of the local area indicating the businesses found.
- On the map indicate where the customers are found.

Support Materials

⇒ Real goods
⇒ Pictures of goods and services
⇒ Pictures of business activities
⇒ Manila paper

Glossary

Commerce	–	buying and selling of goods; it mainly involves trading and activities such as transport, insurance, advertising, finance, warehousing.
Consumer goods	–	these are goods used by the final consumer (powdered soap, rice, and ECCO corned beef)
Legal	–	having to do with the law; a legal service is provided by the lawyer.
Manufacturing	–	make
Market research	–	finding out what people want and need
Producer goods	–	these are goods used by businesses to produce other goods; also called raw materials.

Assessment

- List the names of some businesses and what they do in the community.
- Classify the businesses according to what they do.
- List some goods and services. How many of them are consumer goods?
- Differentiate between goods and services.

UNIT 4.2 PRODUCT DESIGN

Design process

Learners should be able to:

discuss the importance of design.
describe the different stages of a design process.
come up with a variety of possible solutions from a given design brief.
make a product based on the best solution.
test and evaluate the product.

What is designing?

These are objects or items around you that have been made by people to serve particular purposes. For example a chair has been made for people to sit on while clothes are made for people to wear. Before these items were made there was nothing suitable to use. People, therefore, came up with ideas and solutions to solve the problems. They decided

on what shapes, sizes, materials and construction methods they should use. Thinking up solutions to problems is called designing.

Designing or problem-solving involves:

- Identifying a problem or need.
- Thinking about ways of solving the problem.
- Using materials, tools/equipment to make a product or to solve the problem.
- Finding out whether or not the solution is suitable.

When designing an item or artefact, it is important to make sure that the product is:

- safe to use.
- strong enough to carry the expected load or withstand forces exerted on it.
- durable or is able to last a long time.
- attractive.

The importance of design

Sometimes we take for granted the things that are around us without realising that they have been made through careful planning. Think about the chairs you are sitting on, the desks you use for writing, the pencils and pens you do your homework with, the warm clothes that protect you in winter, the shoes that help you to walk long distances. In fact all these things have been created for people to have a comfortable life. So anyone who designs things to satisfy people's needs and improve their lives is a designer.

The design process

This is a set of stages that one has to go through in order to come up with a solution for a practical problem. Below are the important steps to follow when designing:

The problem

A concern that can be identified by you, your teacher, your parents, your friends or some other people.

Research

Finding out about the problem. Gather as much information as possible about the problem. Consult books magazines, friends, teachers or anyone who may be able to help you. Think about ways of solving the problem.

- What the product is for?
- What materials are available?
- How much will it cost?
- How will it be constructed?
- How will it look like?
- How much time you have to do it?

Analysis

Study the information you have gathered. Try to break the problem into smaller parts.

Brief

Write a statement or clear sentence that tells other people what you are going to do.

Ideas and Development

Draw as many possible solutions as you can. Look carefully at these ideas and using the best ones, produce a final solution.

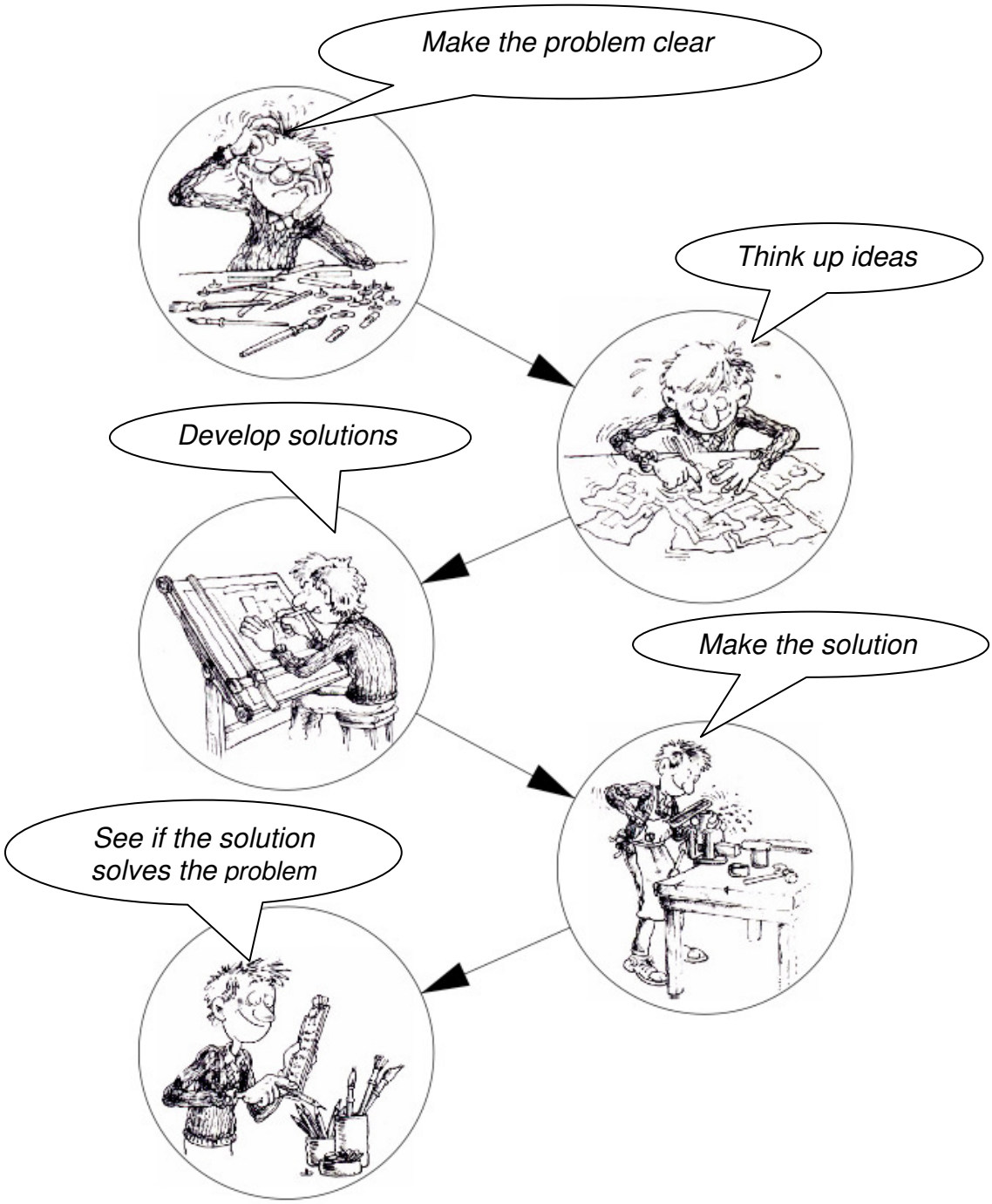
Making

Make your final solution. Carefully plan your work. Use materials, tools and equipment to make the product.

Testing and Evaluation

Test your final work to see if it does what you expected. Say what changes you would make if you had to do it again.

The design process



Suggested Teaching and Learning Activities

- Lead a discussion on the importance of design.
- Explain to the learners that anything that you do not like or makes you uncomfortable is a problem.
- Give them a few examples of designs (artefacts) and ask them in what way they have met people's needs.
- Ask them about their own needs (individual) as oppose to that of the class needs (Group).
- Talk about the important steps to follow when designing. Explain the design process in simple terms. Avoid using the design jargons at this stage.
- Use cartoons and posters to explain the different stages of the design process.

Activity 1 – Design

Scenario

Your friends and relatives always find it difficult to locate your house.

Problem

There is no sign or number to identify your house.

Brief

Design an item that can be used to identify your house easily.

Design considerations:

What numbers are you going to use in your design?

Find out the plot number of your place.
Collect samples of number plates from old magazines.
Select your favourite style. You may decide to come up with your own style.

What will be the appropriate shapes and sizes of the item?

Look at existing number plates from around your area.
Decide on the shapes and sizes of the numbers.
Make sure the numbers can be recognised from a reasonable distance.

What material(s) would you consider for making the item?

Think about suitable materials for making the design.
Consider materials that are available in the school.
Do not use materials that you are not confident in manipulating.

Where and to what material is the item to be fixed?

Think about different ways of fixing the item.
Select a suitable method of fixing the item in place.

What finish will you use on the item?

Think about different finishes that can be used on the item.
Use a suitable finish that will withstand all weather conditions

Activity 2 – Making Processes

How you will make the item? Think about all the different tools and processes that you will need for making the item. These may include:

- **marking out tools** such as pencil, steel rules, compasses, templates and stencils.
- **cutting and shaping tools** such as a coping saw, hand drill, twist drill, smoothing plane, flat, curved and round surforms, hand, half round and round files.
- **holding tools** such as woodwork bench woodworker's vice, G-cramp and bench hook.
- **Joining and forming tools** such as cross pein hammer, pincers, flat or Phillips screwdriver.

Before learners start making their artefact:

- make sure that all the necessary tools, equipment and materials are available in sufficient quantities.
- where necessary, decide on how the learners are going to share tools, equipment, etc.
- be clear about how you are going to show or demonstrate the various processes that the learners are required to carry out to make the items they have designed.

It is therefore important for you to show the learners how to carry out these processes stage by stage. For example, show them how to mark out and allow them to mark out before you show them how to cut.

While they are working move around to provide individual support as needed. Remember it is important to emphasise safe handling and use of materials, tools and equipment. Make constant reference to safety rules.

Activity 3 - Evaluation

After making the product, it is necessary to identify strong and weak points of the product. This involves asking some questions such as:

- Does the product solve the problem?
- Will the product last a long time?
- Is the product attractive?

Support Materials

Make the following craft materials available in sufficient numbers: Jelutong, round head screws, polyvinyl acetate (PVA) acrylic sheets (assorted colours) and tensol cement.

Glossary

Design	-	ways of thinking up solutions to problems.
Force	-	strength or power to hold or withstand the pressure of something.
Artefact	-	an article made by man.
Attractive	-	very pleasing in appearance or causing interest and pleasure.
Scenario	-	a probable or imaginary situation.

Assessment

- Explain the meaning of design?
- What is the importance of design?
- Illustrate the different stages of the design process?
- What is a design brief?

- Why is it necessary to come up with a variety of design solutions when designing?
- Why is evaluation and testing an important part of the design process?

UNIT 4.3 GRAPHIC DESIGN

Postcards and greeting cards

Learners should be able to:

explain graphic design.

identify examples of graphic designs.

use different tools and media to create various effects in postcards/greeting cards.

use the layout technique of letters, words and symbols in postcard/greeting card designing.

design either a postcard or greeting card.

create either a postcard or greeting card using a computer.

Graphics

Graphics is used to mean the visual expressions which are written on a surface. The visual images such as pictures, diagrams, graphs and drawings. Design means the arrangement of elements. The elements are such as shape, line, and colour. Examples of these are posters, road signs, and cartoons.

Layout

Layout is the arrangement of visual elements such as colour, lines, shapes, letters and images on paper to attain a pleasing and interesting design. Elements to be used are to be decided and the order in which are to be presented. The size of the design should be considered and whether the overall design should be vertical, horizontal and or radial.

Layout examples

Radial/circular Horizontal vertical

Postcards/greeting cards

These are designs that uses lettering and pictures to communicate messages and also could be used to pass messages to friends and other people. They could also be displayed in homes. Some contain pictures of people, places and important sayings.

Symbols

A symbol is an object that represents something such as an idea, company or society; for example, a dove is a symbol of peace. These are examples of natural objects that have become symbols of human experience. Visual symbols are used to represent an idea, social groups and clubs.

Suggested Teaching and Learning Activities

- Experiment with different forms of illustration.
- Create sketches to brainstorm for more ideas of what to illustrate.
- Make an illustration of what you have chosen.
- Visually portray stories, songs and poems.
- Create a mural for the school.

Support Materials

⇒ Tins, buckets, rags, papers, magazines, newspapers, pencils, chalk, chalk pastels, coloured pencils, paint-brushes, paints, pallets, ink, scissors, art knifes toothbrushes, glue and rubbers.

Glossary

Lettering - the writing or drawing of letters.
Layout - the division of space required for letters and pictures as well as establishing the margins.
Style - an original way of creating a work of art of one person.
Design - a work of art or the process one follows of arranging parts when creating a work
Format - the flat surface /picture plane (a piece of flat paper or cloth). It can be a rectangle, square and circle. It can also be used in a vertical or horizontal direction.

Assessment

✎ Analyse a design product.

UNIT 4.4 TWO-DIMENSIONAL CRAFTS

Pattern and print making

Learners should be able to:

explain different types of pattern as random and planned.
create patterns using various printing techniques.
design random and planned patterns.
create a composition using any printing technique.
discuss careers relating to pattern and print-making.

Pattern Making

Pattern is a design created by repeating shapes, colours, lines, etc. It could also be made by drawing, painting, printing or spattering paint. It is sometimes used to decorate surfaces. A motif is used to produce beautiful artworks.

The two types of pattern are random and planned pattern. The random pattern is created when shapes or lines etc are repeated not in a given order. In a planned pattern shapes are carefully repeated one after the other. The space too between the shapes has to be equal. There are various kinds of patterns from the natural and man-made environment for example; patterns on clothes, baskets, decorated walls in a form of murals, floor tiles are man-made patterns and those of animals, insects, and plants are natural patterns.

Patterns from the Natural Environment

These are patterns found on insects, animals such as zebras, spots of a guinea fowl, and spider web, (veins in leaves, animal prints in the sand, sand blown by wind).

Patterns from Man-Made Environment

These are patterns found on clothes, wall decorations, bricks of walls, on furniture, utensils etc. (fabric patterns, decoration on ceramic forms, pots, patterns of decorative marks on kitchen objects), baskets, decorated walls, utensils and road markings.

Print Making

It involves the processes of transferring images from one surface to another that could be made by many techniques. A print can be made by pressing an inked or painted object on a surface. Touching surfaces with greasy hands also makes prints. I n art, printmaking is a technique or process of duplicating a design by transferring its image from a prepared surface to another material.

Prints and patterns in the environment and artwork

These include foot prints, leaf prints. finger prints, hand prints, potato prints and block print.

Printing techniques/methods

Introduce a variety of simple printing techniques/methods such as: block printing and screen printing

Processes

Incise the surface of a suitable block, apply paint and print. Pour paint on the board and use a roller to spread the paint evenly over the board, place a hand/finger on the board and transfer on paper.

Natural prints

Natural prints would be prints on soil/sand/muddy surfaces created by animal paws, animal hooves, leaves fallen or dropped on muddy surfaces and footprints.

Man-made prints

These are prints made by pressing an inked or painted object on a surface. These would be prints created by using a block, finger, hand and found objects.

Suggested Teaching and Learning Activities

- Experiment with different ways of making patterns and prints.
- Make sketches of desired patterns and prints.
- Experiment with printing and pattern making techniques.
- Make sketches to brainstorm for more ideas.
- Create a print or pattern.

Support Materials

⇒ The materials are such as paint-brushes, boards, papers, tooth brushes, cardboard's, sticks, containers, nails, paints pallet, pencils, coloured pencils, pastels, chalk, charcoal, ink, scissors and glue, cloth, money clips, art knifes, scissors and found objects.

Glossary

Repetition	-	a series of repeated elements that is similar.
Motif	-	it is a single unit that is used repeatedly to create a pattern.
Planned Pattern	-	it is when the motif is repeated in a well-organised manner.
Random Pattern	-	it is when the motif is scattered all over.
Printing	-	the processes of making images by pressing an inked or painted object on a surface.
Print	-	to make copies of a picture or design or the finished artwork.

Assessment

- Describe prints from the natural and man-made environment.
- Compare and contrast different printing techniques.
- Compare and contrast prints and patterns

- Explain ways of making prints and patterns.
- Identify prints and patterns in the immediate surroundings.
- Analyse / critic the patterns and prints from the natural and man-made environment.

UNIT 4.5 THREE-DIMENSIONAL CRAFTS

Pottery

Learners should be able to:

define pottery.

state sources of clay.

process clay for pottery making.

experiment with pottery techniques of pinch, coil and slab to make forms.

make different forms using the techniques of pinch, slab and coil.

decorate pottery with paint, textures or patterns.

burnish the surface of the leather-hard clay-pot.

In pottery making, methods such as slab, coil and pinch are also used to create pots. Pottery is the making of items from clay. These usually are functional objects such as clay pots, bowls, cups, plates and other clay objects. The clay objects are fired when dry to make them harder and stronger and mot to allow water to pass through. These items we used for various purposes such as storing, cooking and decorating.

Sources of Clay

The clay soil is found in the ground, in termite hills, riverbed and swampy places. The top layer of soil is removed before digging out proper clay.

At a riverbed dried cracked pieces of clay could be found. These pieces could be collected and lose dirt underneath be wiped off and collected

Clay Processing

Clay that is suitable is collected. The natural clay would contain leaves, stones and other impurities. These are to be removed by sieving the dry clay. The dry clay is crushed and screened through a mesh or sieve to remove the lumps such as rocks and impurities. Water is then added to make the clay pliable. This clay has to be stored in a plastic bag or plastic container/bucket.

Pottery Making Tools.

There are different hand building tools such as sticks, nails, small stones, fingers, and buttons. These can be used to draw on clay, make marks and make various textures or decorations on clay forms and shapes to enhance the piece.

Hand building techniques

Coil method

A lump of clay is taken and rolled on a flat surface e.g. table top with a palm of hand, working from the centre outwards. After enough coils have been rolled, they are joined. A few lines are scratched along the edges of the coil to be joined. This helps to join the coils more firmly. More coils are to be joined until the required height is reached or form is made.

If the coils effects are not required, smooth the coil into another on the inside and outside of the form.

Pinch Method
A lump of clay is made into a round or any solid shape. It is then pressed/pinched using the thumb/fingers to form a shape. The clay is gently pulled out until the required shape is formed or made.

Slab Method
A piece of clay is flattened by pressing, patting or using a roller to roll clay back and forth until you get the required thickness. The clay is turned around to avoid it from sticking on the work-surface.

The slab can be textured by rolling clay on a textured surface such as cloth, sack etc. The slab is then cut into the required shape and joined together to make various forms.

Suggested Teaching and Learning Activities
- Collect and prepare clay.
- Experiment with the varied pottery techniques.
- Create various objects using any of the pottery making techniques.
- Finish by decorating the works created.

Support Materials
⇒ Sticks, boards for working on, plastic bags, plastic buckets with lids, cloth or rags, bottle tops and other found objects.

Glossary

Modelling	-	a method of making artwork in which soft material is build up to forms.
Form	-	a three-dimensional object and that encloses volume.
Coil	-	a long round piece of clay used to make artwork.
Slab	-	a method of making forms in which a flat sheet of clay is cut into shapes and joined to make an object.
Pinch	-	a method of making forms in which a thumb is pressed into a piece of soft material to create hollowed out shape.

Assessment
- Analyse pottery works.
- Compare and contrast pottery works.
- Describe methods and techniques used to cerate pottery.

UNIT 4.6 CLOTHING AND TEXTILES

Textile fibres
Learners should be able to:

Describe a fibre.

Classify fibres into natural and man-made.

Classify natural fibres as vegetable and animal.

List examples of vegetable and animal fibres.

List man-made fibres.

Anything that is used to make a cloth is called textiles e.g. clothing, carpets, etc. These textiles come from fibres that are woven and knitted into cloth or fabric. A wide variety of fabrics is available such as cotton, wool, nylon, poplin, etc.

Definition of a fibre

A fibre is a single thread and smallest unit from which fabrics are made from. Fibres can be classified into natural and man-made.

Classification of fibres

Fibres can be classified into natural and manmade fibres. **Natural fibres** are made from materials that are from nature, for example, cotton, wool, silk, linen, etc. **Man-made fibres** are made from mixing different chemicals by in a factory, for example, nylon, acrylic, polyester, etc.

Classification of natural fibres

Natural fibres come from plants and animals. So they are:

Plant/Vegetable fibres

These are from **cotton** is a fibre obtained from the ripe pods of the cotton plant. Cotton is sot and absorbent. Cotton creases easily, that is why you have to iron it. Frequent washing and exposure to the sun discolour cotton fabrics. Linen, sisal, etc. and goat,

Animal fibres

These come from **wool, hair**: Wool comes from merino sheep and angora and hair from animals such as hair, camel. Wool is durable and very expensive, and also hairs are very expensive. **Silk:** Any fibre that is produced by the silk worm is called silk. Silk is very durable. It also reflects light and it does not need ironing. Silk is a very expensive fabric. Silk garments are normally dry-cleaned.

Mineral fibres

These come from asbestos.

Suggested Teaching And Learning Activities

Feeling test

- Examine the different pieces of fabrics displayed, feel each piece and discuss how it feels.
- Group pieces of fabric according to how it feels.
- Learners should feel their own clothing and compare with the pieces of fabric.

Burning test

- Have pieces of cotton and woollen fabrics
- Cut a piece of each
- Burn and smell each piece.
- How does it smell like?
- What is the colour of the residue?

N.B. All the tests should be done by learners under teacher's supervision.

Support Materials

⇒ Samples of different fabrics

⇒ Samples of fibres
⇒ Dressmakers cutting scissors
⇒ Matches
⇒ Enamel plate to collect the residues

Glossary
Fibre - A single thread and smallest unit from which fabrics are made.
Textiles - Anything that is used to make cloth.

Assessment
- Define a fibre
- Classify fibres
- Give examples of natural fibres
- Give examples of manmade fibres

Sewing tools and basic hand stitches

Learners should be able to:

Identify basic sewing tools.

Classify basic sewing tools according to use.

Discuss care and storage of sewing tools.

Classify basic hand stitches and their uses.

Practise basic hand stitches of tacking, back stitch running and hemming.

Apply basic hand stitches to a project.

A sewing project is easier to make with the help of a sewing tool. Knowing which tool to select and where to use it will simplify and improve the outcome of almost every sewing step you encounter.

Sewing tools

Sewing tools make tasks easier, improve results and certainly add to sewing enjoyment. However, there are many sewing tools available today. You need to choose the tools that will be of use to you. A good tool is the one which is easy to use and is the one that I use quiet often For example sewing a dress, shirt or even a hem. There are different tools used for sewing. These tools are classified into:

Stitching tools
These tools are used for general sewing and embroidery, for example sewing needles.

Cutting tools
These tools are used for cutting fabrics and paper, for example paper scissors.

Ironing tools
These tools are used for removing creases in our clothing and articles, for example an iron.

Marking tools
These tools are used for making a mark on the fabric. For example; one can use tailor's chalk to mark where one is going to stitch.

Care and storage of sewing tools

Different tools are cared for differently. Make sure you protect the tools from dust and water. Store them neatly when not in use. You can store them in a cupboard or box.

Basic hand stitches

Basic hand stitches can be divided into four groups:
***Temporary*:** e.g. tacking (even, long and short), diagonal tacking, tailor's tacks, etc.
Joining: Running, backstitch, over sewing, machining, etc.
Neatening: Hemming, loop stitch, buttonhole, herringbone, overcastting, machining, etc.
Decorative/embroidery: Chain stitch, satin stitch, stems stitch, French knots, etc.

Suggested Teaching and Learning Activities

- Practice the use of each sewing tool available.
- Store the sewing tools correctly.
- Look at the different tools on display and classify them.
- Visit a local dressmaker or designer or a store. Make a list of equipment found there and find out the cost of each item.
- Make samples of the different basic stitches
- Draw the different stitches and label them
- Print your name on a piece of cloth and illustrate how the stitches might be used.

Support Materials

⇒ Sewing tools
⇒ Flash cards
⇒ Posters of sewing tools
⇒ Samples of different stitches

Glossary

Temporary stitch	-	used for a short period of time and then removed
Joining stitch	-	joins two or more pieces of fabric together permanently
Neatening stitch	-	Finishes the raw edges to avoid ravelling
Decorative stitch	-	used to decorate a piece of fabric

Assessment

- Group sewing tools according to their classes.
- Give two examples of temporary stitches and two examples of neatening stitches.
- differentiate between (i) temporary stitch and (ii) neatening stitch.

Macramé

Learners should be able to:

Explain macramé.

Discuss the use of macramé.

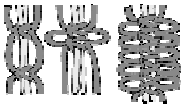
Demonstrate techniques of macramé.

Apply techniques of macramé to a project.

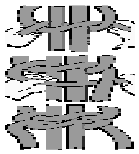
Macramé is a method of knotting or weaving a string or yarn together to make ornamentals. It is the craft of tying or knotting cords in geometric patterns to create jewelry, plant holders, home decorations, and many other creative crafts. Any thick thread can be used for macramé e.g. wool, embroidery thread, string, rope, etc.

In macramé there are only two main knots to make, namely square knot and half hitch. From the two knots you can come up with new ways of combining knots.

Square knot and variations



Square knot



Half square Knot

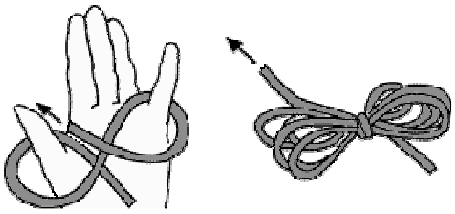


Alternate square knot

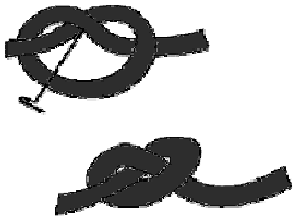


Half hitch

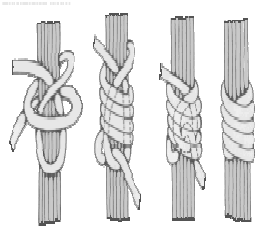
Other important knots



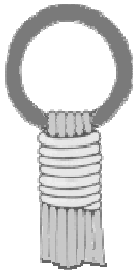
Bundle cords



Overhead knot



Gathering wrap



Mounting cords in a ring

Suggested Teaching and Learning Activities

- Practise variation of square knots.
- Practise variation of half hitch.
- Make a project using hitch and square knots.

Support Materials

⇒ Cords of different thicknesses e.g. wool, string, embroidery threads, rope, etc.
⇒ Mounting rings, sticks, rods, frames, cable, etc.

Glossary

Half hitch	-	a knot tied to a post, cable or ring
Macramé	-	the craft of tying or knotting cords in geometric patterns
Square knot	-	a reef knot

Assessment

- Explain macramé
- List items that can be made from macramé
- Give the two main knots used in macramé

Embroidery and decorative stitches

Learners should be able to:

List embroidery stitches.
List the uses of embroidery stitches.
Practise creative ways of using embroidery hand stitches.
Apply embroidery stitches to a project.

Embroidery stitches are used to decorate fabrics. Different colours of sewing threads or embroidery threads and embroidery stitches can be used. Examples of embroidery stitches are: chain stitch, cross-stitch, herringbone stitch, lazy daisy and satin stitch.

Suggested Teaching And Learning Activities

- Practise different embroidery stitches.
- Make a project using embroidery stitches.

Support Materials

⇒ Samples of articles with embroidery
⇒ Assorted colours of embroidery threads
⇒ Assorted colours of embroidery threads
⇒ Embroidery frames, embroidery hoops
⇒ Cross-stitch fabric, any fabric suitable for embroidery

Glossary

Embroidery - decorative needlework done usually on loosely woven cloth or canvas, often being on a picture or pattern

Assessment

- List embroidery stitches
- List different threads that can be used for embroidery
- Explain embroidery in your own words

Textile construction processes

Learners should be able to:

Explain seams.
Identify types of seams.
Make samples of different seams.
Apply seams to a project.

Seams

A seam is made when you join two or move pieces of fabrics together. There are different types of seams that one can use. For example, open seam, French seam and double stitched seam. Different seams can be used for different articles. One can use more than

one seam in one garment. The seam which is common in children clothes is the French seam because it is strong and self neatened.

Suggested Teaching and Learning Activities

- Make samples of the open seam, French seam and double stitched seam.
- Use different seams in a project.

Support Materials

⇒ Sewing tools
⇒ Articles with different seams
⇒ Assorted fabrics of different qualities
⇒ Assorted sewing threads

Glossary

Seam - joining of one or more pieces of cloth together using a line of stitching

Assessment

- What is a seam?
- List two examples of seams
- Where can an open seam be used

UNIT 4.7 TECHNOLOGY

Structures

Learners should be able to:

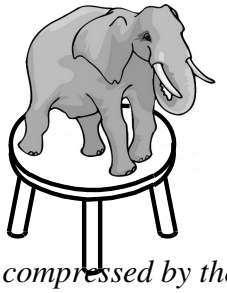
explain forces.
differentiate between static forces and dynamic forces.
use the principle of triangulation to achieve stability and rigidity.
experiment with materials such as card, straws, dowels, wood strips, metal strips and others to make model structures.
design and make model structures to solve specific problems.
test model structures

Forces

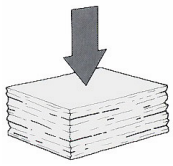
A force is something that makes an object move. There are different ways of applying force to an object. These include compression, tension, torsion, bending and shear.

Compression

A force that tries to press or squash an object is referred as a **compressive force**.

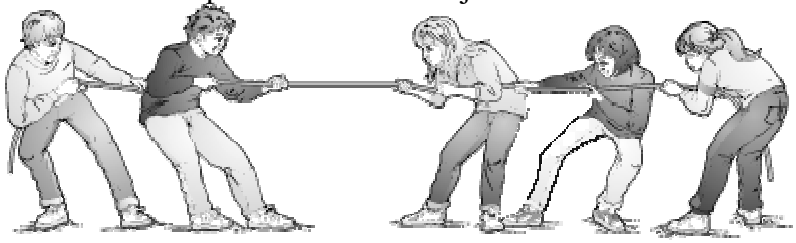


The stool is being compressed by the weight of the elephant.



Compression is squashing

Tension
A force that tries to pull or stretch an object is referred as a **tensile force**.

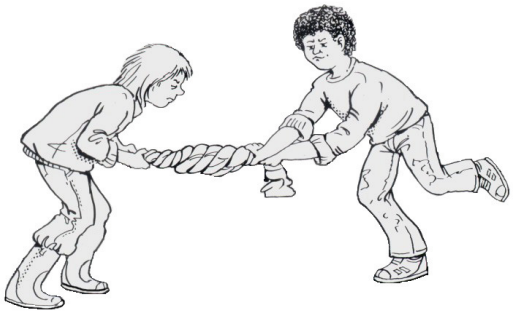


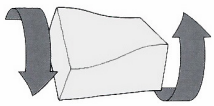


Tension is pulling apart

The rope is in tension as it is being pulled in opposite directions.

Torsion
A force that tries to twist an object is referred as a **torsion force**.

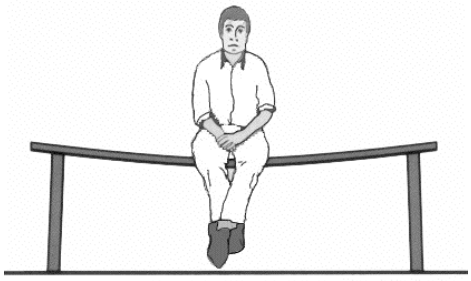


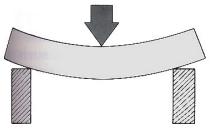


Torsion is twisting

The wet towel is being twisted to squeeze out the water.

Bending
A force that tries to bend an object is referred as a **bending force**.

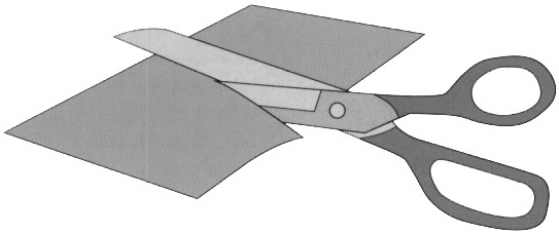


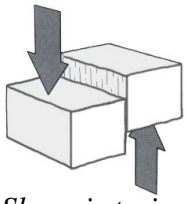


Bending is curving

The bench is bending under the load (weight) of the man.

Shear
A force that tries to cut across an object is referred as a **shear force**.





Shear is trying to slice

To cut paper, the blades of the scissors are forcing the paper to break, by forcing one side of the cut upwards and the other side downwards. The paper is being put into shear.

How forces act on frame structures

A force is anything that changes the speed, direction and movement of an object. When you push a wheelbarrow to move it, the push is a force. When you pull a door to open it, the pull is a force. When you stand on an empty cardboard box, the force you exert bends or flattens the box. But not all forces have apparent effect. For instant, when you sit on a chair, your body weight pushing down on the chair is a force. Below are some examples of forces acting on frame structures.

Compression



The chair's legs are squashed, or in compression.

Tension



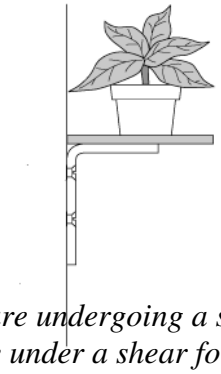
The ropes of the swing are pulled, or in tension.

Torsion



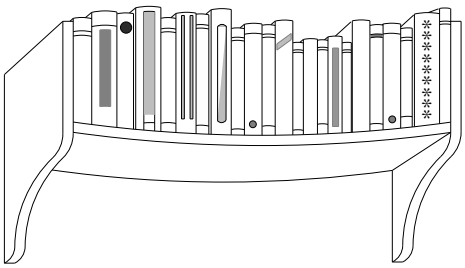
The screwdriver tightening a screw is subjected to torsional load, or is under torsion.

Shear



The screws are undergoing a shearing action or are under a shear force.

Bending



The bookshelf is bending in the middle due to the heavy weight of the books. The load acting on the shelf is a bending force.

Static force

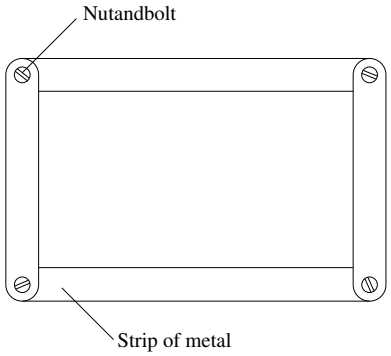
A stationary force applied to a structure. E.g. a person sitting still on a chair is a static force.

Dynamic force

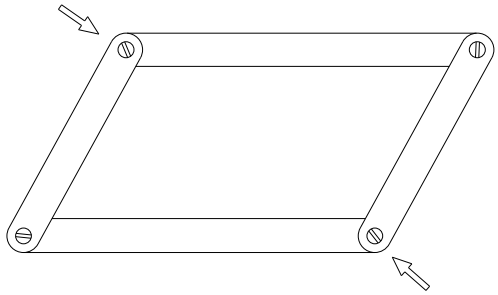
A moving or changing force applied to a structure. E.g. a person bouncing on a trampoline.

Making structures rigid - Experiment 1 (Two dimensional frames)

Join the strips of metal using bolts and nuts to form a rectangular frame as shown in the diagram below.

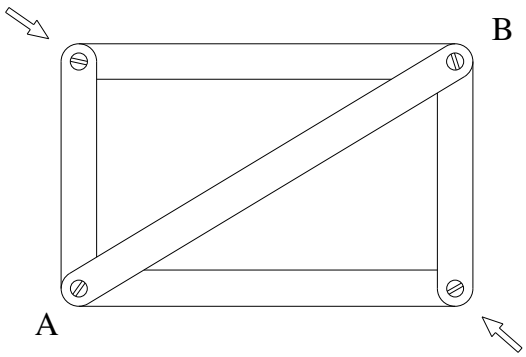


Apply force to the rectangular frame and see what happens.



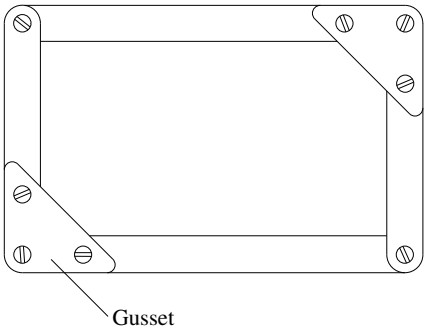
It becomes out of shape easily. Hence we say that the frame is not rigid.

Add an additional strip to the frame at A and B. Try to push or pull the frame to see if there is any change.



The additional strip prevents the corners A and B from moving apart. The frame can no longer be forced out of shape. We therefore say the frame is rigid. Note that the additional strip has formed two triangles. The effect is known as triangulation.

The use of gusset plates as shown below can also add rigidity to a frame structure.



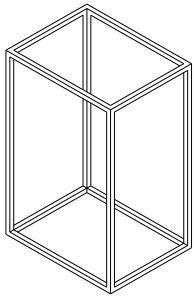
A gusset is a piece of material used to join the members of a structure. It comes in different shapes and sizes. The ones used in the structure above is a triangular plate.

Triangulation

A triangle is the most rigid structure. The process of joining strips of material together to form triangles is called triangulation. The principle of triangulation is used in many structures. Examples include gates, roofs, bridges, beams, pylons, furniture and cranes.

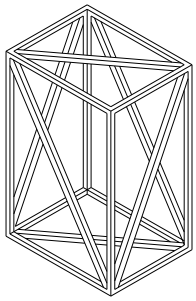
Making structures rigid - Experiment 2 (Three dimensional frames)

The framework made up of square tubes shown below can easily be distorted when pressure is applied at the corners. How can this be prevented?



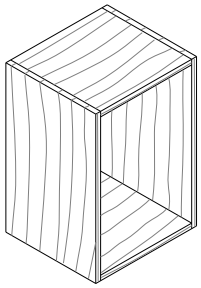
The framework is not rigid.

By applying the principle of triangulation i.e. adding more tubes to form triangles the framework will not be distorted.



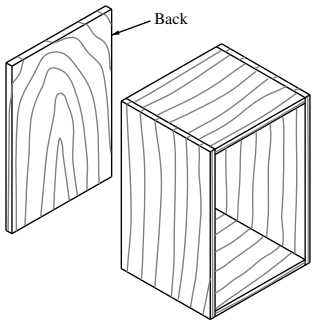
The framework is now rigid.

An open box shown below will tend to distort. How can this be prevented?



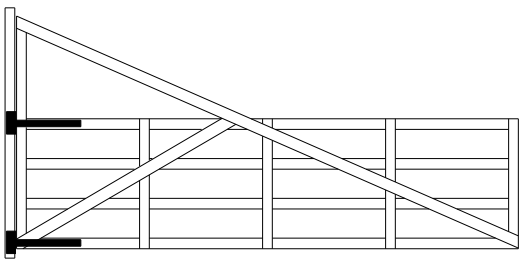
The box can be easily distorted.

Adding a back part will prevent the corner joints from twisting.

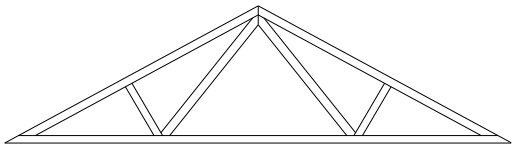


The box will not be distorted.

Examples of triangulation



Large gate

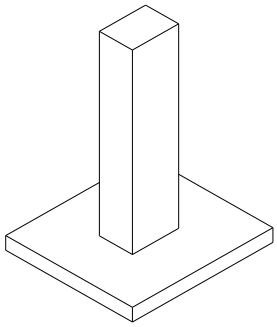
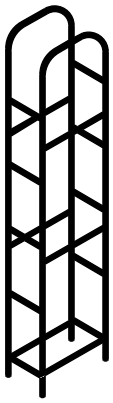
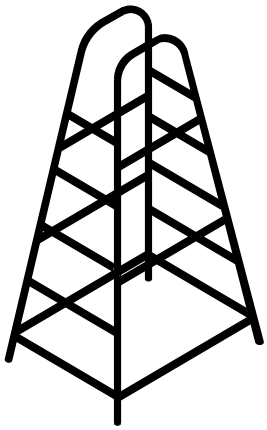


Roof truss

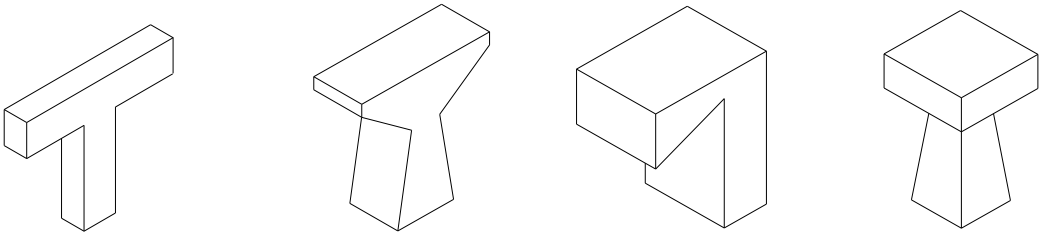
Ask learners to look for other examples of triangulation from old magazines and paste them in their notebook.

Making structures stable - Experiment 3

Structures need not only be rigid but also stable so that they do not fall over easily.

The rectangular block place on its end can easily fall over with a slight push. How can this be prevented?  *The block is not stable.*	By joining a large base to the end of the block.  *The block is now stable.*
Similarly the frame below is likely to topple easily. How can this be prevented?  *The frame is not stable.*	By making the base bigger than the top.  *The frame is now stable.*

Experiment with the different forms shown below to see which ones are stable.



(a) **(b)** **(c)** **(d)**

Suggested Teaching and Learning Activities

- Explain to learners the different types of forces mentioned above.
- Use pictures or diagrams to illustrate each of the forces.
- Get learners to demonstrate the practical applications of the different forces.
- Explain to learners the concept of triangulation.
- Demonstrate the practical application of triangulation.
- Illustrate different forms of stable and unstable structures.

Activity 1: Demonstrate understanding of forces

Demonstrate practical examples of the types of forces using learners' immediate surroundings. Ask learners to demonstrate practical examples of each type.

Activity 2: Identification of the types of forces

Assign learners to cut pictures of examples of each type of forces from old magazines and paste them in their notebooks.

Activity 3: Application of forces

Through experiment demonstrate the application and usefulness of each type of forces so that learners are able to apply them.

Activity 4: Triangulation

Demonstrate practical examples of triangulation using learners' immediate surroundings. Ask learners to come up with their own solutions to practical problems of making structures rigid.

Activity 5: Stability

Through experiment demonstrate the importance of stability in structures. Assign learners to sort stable and unstable structures from a variety of forms.

Support Materials

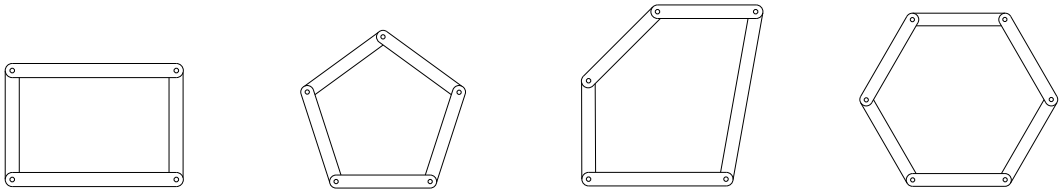
- ⇒ Cut outs or pictures of examples of forces
- ⇒ Building blocks such as Lego kit or wooden blocks
- ⇒ Materials - cardboards, straws, dowels, paper glue, wood glue, wood and metal strips
- ⇒ Fixings - staples, paper clips, paper fasteners, nuts and bolts
- ⇒ Tools - staplers, hole punches, revolving punch pliers, coping saws, junior hacksaws, hand drills, drill bits, flat or Phillips screwdrivers, trimming knives, combination pliers and glue gun

Glossary

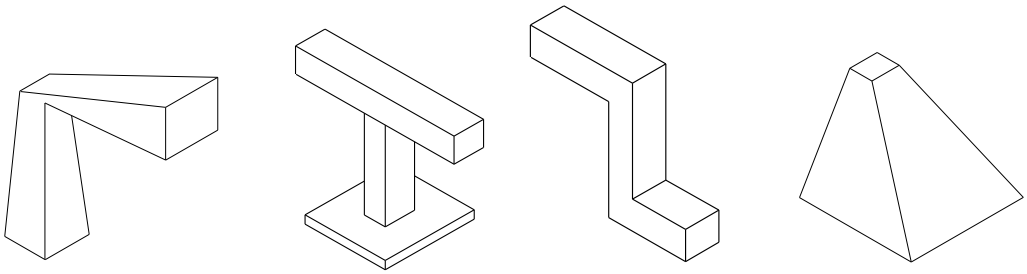
Force	-	the power of moving an object. A pull or push action.
Load	-	the amount of weight carried by a structure.
Dynamic force	-	a moving destructive force.
Rigid	-	not easily changed or moved into a different shape or position.
Stable	-	a structure which will not fall over easily when a force is applied.
Static force	-	A force at rest, still.
Structure	-	anything that can support a load or itself.
Triangulation	-	The process of joining strips of material together to form triangles

Assessment

- What is a force?
- Illustrate the different types of forces.
- Differentiate between the different types of forces.
- Name and draw the symbols of the different types of forces.
- What is a common word for compression?
- What type of force is acting on the legs of a table?
- What parts of a child's swing are in tension?
- Name one part of a bicycle which is torsion when it is being ridden.
- Use the minimum number of strips to make the following shapes rigid:



- Give examples of structures where the principle of triangulation has been used.
- Classify the following forms into stable and unstable structures.



Mechanisms

Learners should be able to:

illustrate the four basic types of motion.
identify crank mechanisms.
identify the input and output of a crank mechanism.
demonstrate the mechanical advantage of crank mechanisms.
use crank mechanisms to solve specific problems.

Mechanisms or machines make work easier by controlling movement. It helps people do things faster and with less effort. For example, it is easier to push grocery in a trolley than to carry grocery in a big bag.



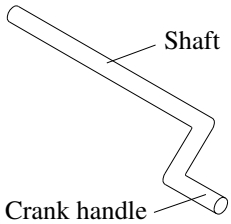
Types of motion

There are four types of motion or movement – linear, rotary, oscillating and reciprocating.

Linear motion is movement in a straight line.
Rotary motion is movement in a circular motion around a pivot.
Oscillating is movement swinging backwards and forwards around a pivot.
Reciprocating is movement moving backwards and forwards in a straight line.

Crank mechanisms

A crank is a device consisting of an arm that is connected to a shaft at right angles; enable the transmission of motion to or from the shaft. They are often attached to handles and used to make turning and tightening easier.



Examples of crank mechanisms



Pencil sharpener
revolving handle



Tricycle pedals



Eggbeater
revolving handle



Carpenter's brace handle



Meat mincer handle

Suggested Teaching and Learning Activities

- Explain to learners how the crank mechanisms make the transmission of motion easier.
- Use pictures or diagrams to illustrate crank mechanisms.
- Explain to learners the inputs and outputs of crank mechanisms.
- Demonstrate the mechanical advantage of crank mechanisms.
- Assign learners to solve specific problems of crank mechanisms.

Activity 1: Demonstrate understanding of crank mechanisms

Demonstrate practical examples of crank mechanisms using learners' immediate surroundings. Ask learners to come up with their own examples.

Activity 2: Identification of crank mechanisms

Assign learners to cut pictures of examples of crank mechanisms from old magazines and paste them in their notebooks. Ask learners to identify the inputs and outputs of crank mechanisms.

Activity 3: Application of crank mechanisms

Through experiment demonstrate the mechanical advantage of crank mechanisms so that learners are able to apply them.

Support Materials

- Cut outs or pictures of examples of crank mechanisms
- "Lego" kit on mechanism or "Mecano" kit for demonstration
- Materials - cardboards, straws, dowels, paper glue, wood glue, wire, wood and metal strips
- Fixings - staples, paper clips, paper fasteners, screws, nuts and bolts
- Tools - staplers, hole punches, revolving punch pliers, coping saws, junior hacksaws, hand drills, drill bits, flat or Phillips screwdrivers, trimming knives, combination pliers and glue gun

Glossary

Crank	-	a device consisting of an arm or handle that is connecting to a shaft or handle at right angles.
Input	-	the effort applied to a mechanism in order to do work.
Mechanical advantage	-	allows the user to move a large load without needing a large force.
Output	-	the total work done by a mechanism.
Shaft	-	a rotating rod that provides motion or power for a machine.

Assessment

- What is a crank mechanism?
- Identify the inputs and outputs of crank mechanisms?
- Explain the mechanical advantage of a crank mechanism.
- Design and make a model of a crank mechanism that will allow people to draw water from a well.

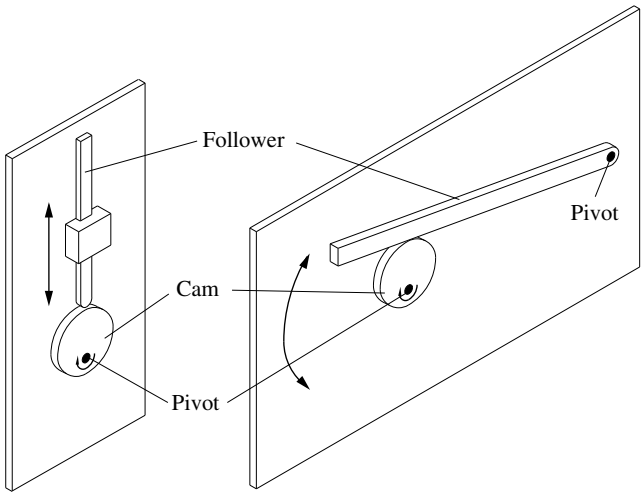
Cam and follower

Learners should be able to:

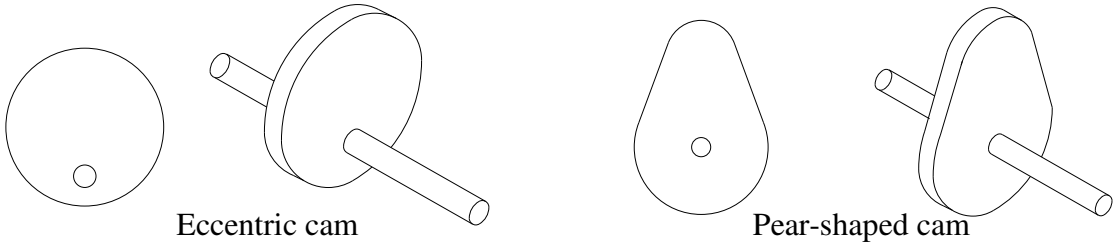
identify cam and follower mechanisms.
identify the input and output of a cam and follower mechanism.
illustrate different types of cams.
demonstrate the different movements of cam and follower mechanisms.
use the cam and follower mechanisms to solve specific problems.

Cam and follower mechanisms

A cam is a mechanism that changes rotary motion into reciprocating motion (up and down movement) or oscillating motion (swinging movement) in a follower.



Types of cams



Eccentric cam

Pear-shaped cam

Suggested Teaching And Learning Activities

- Explain to learners the different types of motion.
- Use pictures or diagrams to illustrate the types of motion.
- Explain to learners the inputs and outputs of cam and follower mechanisms.
- Demonstrate the use and application of the cam and follower mechanisms using practical examples.
- Assign learners to solve specific problems of cam and follower mechanisms.

Activity 1: Demonstrate understanding of cam and follower mechanisms
Demonstrate practical examples of cam and follower mechanisms using learners' immediate surroundings. Ask learners to come up with their own examples.

Activity 2: Identification of cam and follower mechanisms
Assign learners to cut pictures of examples of cam and follower mechanisms from old magazines and paste them in their notebooks. Ask learners to identify the inputs and outputs of cam and follower mechanisms.

Activity 3: Application of cam and follower mechanisms
Demonstrate the different movements of cam and follower mechanisms so that learners are able to apply them.

Support Materials
⇒ Cut outs or pictures of examples of cam and follower mechanisms
⇒ "Lego" kit on mechanism or "Mecano" kit for demonstration
⇒ Materials - cardboards, straws, dowels, paper glue, wood glue, wire, wood and metal strips
⇒ Fixings - staples, paper clips, paper fasteners, screws, nuts and bolts
⇒ Tools - staplers, hole punches, revolving punch pliers, coping saws, junior hacksaws, hand drills, drill bits, flat or Phillips screwdrivers, trimming knives, combination pliers and glue gun

Glossary

Axis	-	centre of rotation
Cam	-	a shaped disk or eccentric wheel
Eccentric cam	-	a circular cam with offset axis
Follower	-	a slider which rubs on the outside edge of a cam
Linear	-	a straight line
Motion	-	a movement or action
Offset	-	not in the middle
Oscillating	-	movement back and forth along an arc
Pivot	-	point around which an object turns
Reciprocating	-	movement backwards and forwards along a straight line
Rotary	-	circular movement

Assessment
✎ What is a cam and follower mechanism?
✎ Identify the inputs and outputs of cam and follower mechanisms?
✎ Illustrate the four types of motion.
✎ Design and make a pull along toy that uses a cam to achieve interesting effects.

UNIT 4.8 MATERIALS AND MAKING PROCESSES

Craft materials

Learners should be able to:
name craft materials.
describe properties and characteristics of craft materials.
list the different uses of craft materials.

Craft materials				
Name	**Characteristics**	**Advantages**	**Disadvantages**	**Uses**
Jelutong	*A soft, lustrous, fine, even textured wood with straight grain. It creamy pale brown in colour and light in weight.*	*Strong and easy to work. It glues and vanishes well.*	*Can be full of resin. It is very expensive.*	*Pattern making, drawing boards, carving and small furniture.*
Acrylic	*Hard wearing, durable and available in sheets or sections. It is a good electrical insulator.*	*Stiff and hard. It bonds well with adhesives such as tensol cement. It can be bent, formed or twisted.*	*Scratches and splinters easily.*	*Light units such as laminated signs, watch glasses, car rear light units, baths and wash basins.*
Panel pin	*A round shank and a small head. It is made from bright mild steel.*	*The head is easily punched below the surface of the wood.*	*Difficult to remove without damaging the surface of the wood.*	*For securing thin plywood or hard board and for fixing small joints.*
Round wire nail	*Also known as French wire nail. It has a round shank and a flat head. It is made from bright mild steel.*	*It gives a strong fixing.*	*It leaves an untidy finish, as the head cannot be punched below the surface of the wood.*	*It is used for general construction work such as roofing, framing and case making.*
Round head screw	*A round head and flat under side. It made from mild steel and is sometimes coated with brass.*	*The round head has a domed top and is quite decorative. It gives a strong fixing.*	*It does not have a strong grip when used in end grains.*	*It is mainly used for securing metal fittings such as handles and nameplates to wood.*
Polyvinyl acetate (PVA)	*Most common type of wood glue. It is white creamy and comes in ready to use state.*	*PVA gives a very strong joint. It is available for both indoor and outdoor use.*	*Extremely difficult to remove excess glue once it has set.*	*It mainly used for gluing wood. Can also be used on textiles and leather.*
Tensol cement	*It is an acrylic based and comes ready to use state.*	*It bonds well with acrylic surfaces and dries very quickly.*	*The fumes given off are very unpleasant. The glue must be used in a well-ventilated area.*	*It is mainly used to glue acrylic to acrylic.*

Measuring and marking out tools

Learners should be able to:

recognise measuring and marking out tools.
use measuring and marking out tools correctly.
realise the importance of using datum surfaces and datum points.

Ruler



A ruler is the most important equipment in the school workshop. It is used to accurately measure and mark out lenghts, widths or depths and thicknesses.

Pencil



A pencil is used for marking wood.

Felt pen



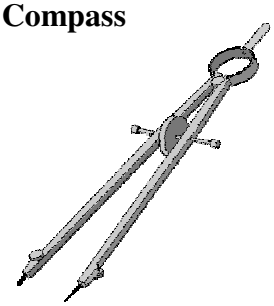
A felt pen is used for marking acrylic/ plastic.

Scriber



A scriber is used for marking metal and plastic.

Compass



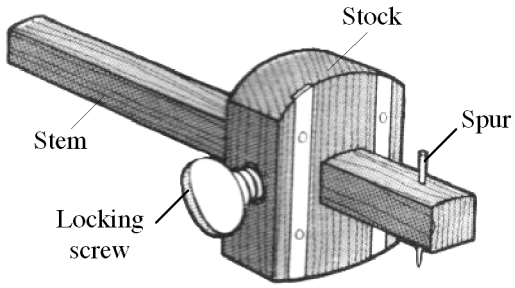
A compass is used for drawing arcs and circles on materials

Try square



A try square is used to mark and check 90° angles

Marking gauge



A marking gauge is used to mark out the width and thickness of materials.

Suggested Teaching and Learning Activities

- Demonstrate the correct methods of using measuring and marking out tools.
- Let learners practice the skills of measuring and marking out.
- Explain the importance of measuring and marking out materials accurately.
- Ask learners to identify the measuring and marking out tools by names.

Support Materials

⇒ Charts illustrating the correct methods of using measuring and marking out tools
⇒ Pencils, scribers, felt pens, compasses, try squares and marking gauges

Glossary

Datum line- a reference line on a flat surface from which all measurements can be taken, e.g. face side and face edge.

Assessment

- Using a ruler, measure the following lengths: 65 mm, 78 mm, 157 mm and 292 mm.
- Draw and label a try square.
- What is the use of a marking gauge?
- Explain datum point and datum line.
- Why is accurate marking out necessary when making products?
- Which tool should be used to test the squareness of a piece of work?

Holding tools

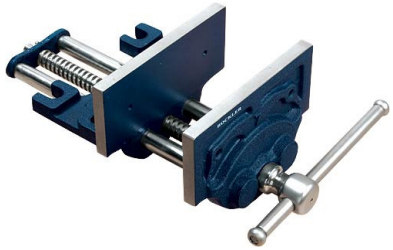
Learners should be able to:

identify holding tools.

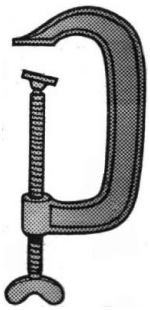
use holding tools appropriately when working with wood and acrylic.



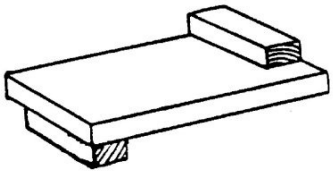
Workbench



Woodworker's vice



G-cramp



Bench hook

Workbench
The workbench is an important piece of equipment in a workshop. Its main purpose is to support the material being worked. It should always be kept clean and tidy.

Woodworker's vice
The woodworker's vice is fixed to the workbench. Wooden jaws are normally fitted to avoid markings. It is mainly used for holding work.

G-cramp
A G-cramp is used to hold work onto the workbench. It can also used to clamp small pieces of glued wood together.

Bench hook
A bench hook is used for holding a piece of wood on the bench when sawing.

Suggested Teaching and Learning Activities
- Demonstrate the different ways of holding work onto the workbench.
- Ask learners to apply these methods when working on their products.

Support Materials
⇒ Charts illustrating the different ways of holding work on the workbench
⇒ Workbenches, woodworker's vices, G-cramps, bench hooks, acrylic, and jelutong.

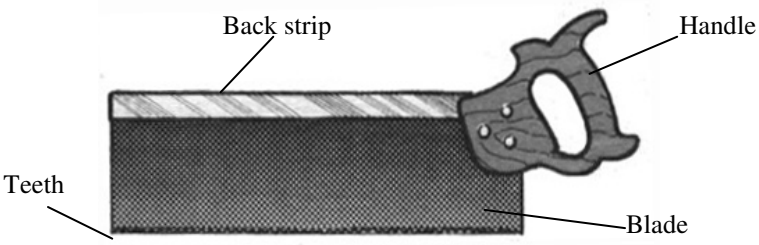
Glossary
Acrylic - A brittle plastic commonly known as Perspex. It is available in a number of colours

Assessment
- Why should you use a bench hook when cutting across a piece of wood?
- How do you stop the jaws of a vice from marking soft surfaces?
- Which tool should be used to test the squareness of a piece of work?
- Sketch a G-cramp and a bench hook.

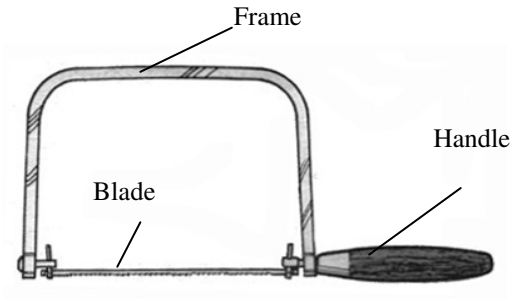
Cutting and shaping tools

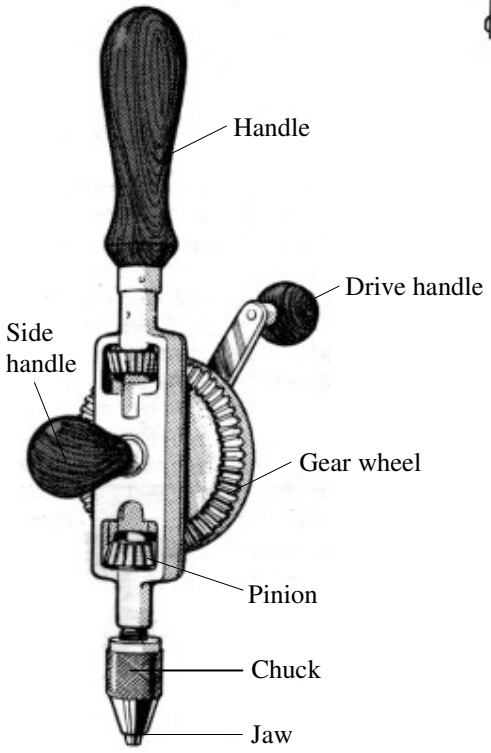
Learners should be able to:

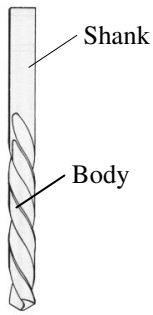
list cutting tools.
cut wood and acrylic to the required sizes.
plane the edges of wood and acrylic to the required sizes.
shape wood and acrylic to specified requirements.
drill through holes and pilot holes in wood.
drill through holes in acrylic.



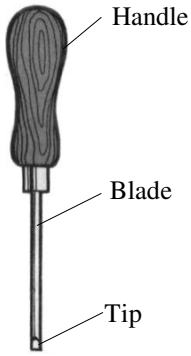
Tenon saw



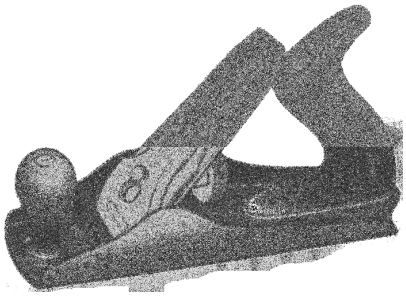




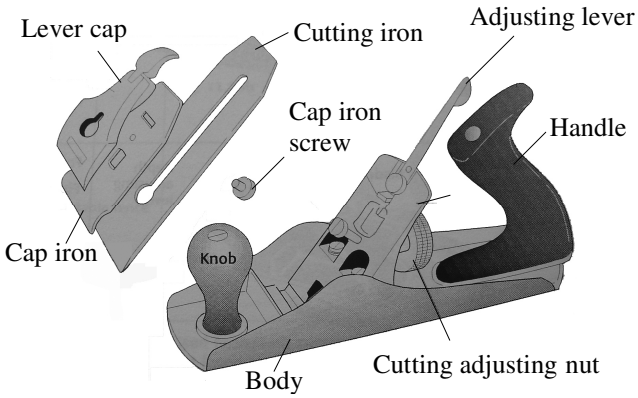
Twist drill



Bradawl



Smoothing plane



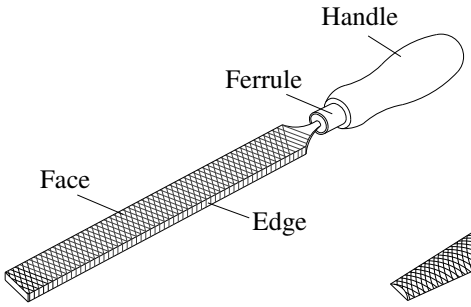
Parts of a smoothing plane



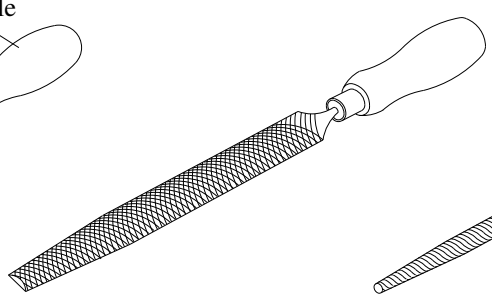
Flat surform



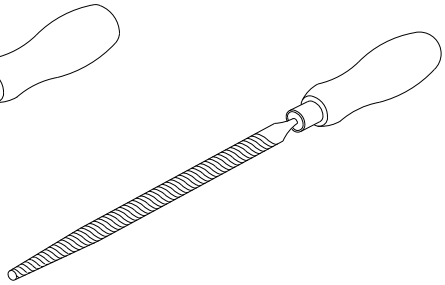
Round surform



Flat file



Half round file



Round file

Tenon saw
The tenon saw is used for light bench work and for cutting joints.

Coping saw
A coping saw is used to cut wood and some plastics. It is also suitable for cutting intricate shapes. It cuts with a backward stroke, so it is important that the blade is placed with the teeth pointing towards the handle.

Hand drill
The hand drill is a handy too for drilling holes in materials. The chuck can take up to 8mm twist drill. The hand drill is operated by cranking the drive handle.

Twist drill
The twist drill is used for drilling holes in materials such as wood, metal and plastic. It is usually available from 3mm to 12mm.

Bradawl
A bradawl has cylindrical blade and sharp tip. It is used for making small holes for nails and screws in wood and for piercing holes in leather.

Smoothing plane
Smoothing plane is used for smoothing wood and edges of acrylic. The curly waste produced when planning is called shavings.

Surform tools
Surform tools are used for shaping wood. Each tooth has a hole through the blade to help clear the waste and stopped clogging. They are available in different shapes (flat, curved and round) and lengths.

Files
Files are used to shape and smooth materials including wood metal and plastic. They come in different shapes, cuts and grades. Some files come with a safe edge, which stops them cutting adjacent to the one being filed. They are also available in a variety of sections – flat, half round and round.

Suggested Teaching and Learning Activities

- The teacher should demonstrate the techniques of:
  - straight cutting using a tenon saw.
  - curved cutting using a coping saw
  - drilling using a hand drill.
  - smoothing a piece of wood using the smoothing plane.
  - shaping a piece of wood using surform tools
  - shaping a piece of acrylic using files
- Ask learners to practice and apply the techniques mentioned above when working on their products.

Support Materials

⇒ Illustrating the techniques of straight cutting, curved cutting, drilling, planning, removing waste with surform tools as well as filing
⇒ Workbenches, woodworker's vices, G-cramps and bench hooks
⇒ Tenon saws, coping saws, hand drills, a set of twist drills and bradawls
⇒ Smoothing planes, flat surforms, round surforms, flat files, half round files and round files
⇒ Acrylic and Jelutong

Glossary

Handy	-	useful and easy to use
Intricate	-	containing many details, complicated and complex
Stroke	-	a single complete movement of the arms
Technique	-	the procedure, skill or art used in a particular task

Assessment

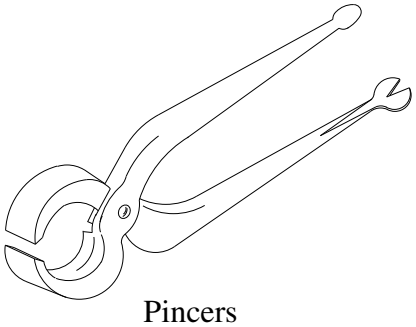
- Why should you clamp materials when drilling?
- Why are soft jaws required when holding plastics in a vice?
- Why are surforms always used when removing waste on wood?
- Sketch a bench hook.
- Label the parts of a coping saw.
- What is the use of the bradawl?

Joining and forming tools

Learners should be able to:

recognise joining tools.

join wood using different methods including dovetail and staggered nailing, round head screws, bolts and nuts, polyvinyl acetate (PVA) and tensol cement.



Pincers



Flat screwdriver

Cross pein hammer

A cross pein hammer is used for light work such as nailing small nails into wood. The pein is used for starting small nails.

Pincers

Pincers are used for removing small nails. The claw at the end of the handle is for lifting bent nails.

Flat screwdriver

Flat screwdriver has parallel or ground back flared tip and is used for inserting and removing screws with straight slots.

Philips screwdriver

Philips screwdriver is used for inserting and removing crossed head slot screws or Philips types screws

Ring spanner

A ring spanner is used for tightening or loosening nuts and bolts.

Finishing

Learners should be able to:

identify different types of finishes.
prepare surfaces of wood using appropriate abrasives and clean up techniques.
prepare edges of acrylic using appropriate abrasives and clean up techniques.
follow the stages of applying wax finish to a wooden product.

Types of finishes

There are many different types of finishes that can be used on wood, metal, plastic and other materials. These include paint, wax and varnish. For wooden projects only wax polish has been suggested for this level.

Surface preparation

In the process of making products, materials are subjected to a variety of operations such as cutting and drilling. These operations leave behind marks and rough surfaces, thereby spoiling the beauty of the product. Also such marks may hurt people who use the product. To remove marks and smooth edges and surfaces, we use abrasives.

Types of abrasives

There are different grades of abrasives, ranging from very coarse to extra fine. In choosing a grade of abrasive, you have to take into account the condition of the surface on which you are working and the degree of smoothness required. If necessary, start with a coarse grade that removes larger marks quickly and easily, then gradually work down to the finer grades of abrasives for a final finish.

Finishing wood

Before applying a finish, you must ensure that your surface is well prepared. This should include the following:

- Punch all nails below the surface.
- Smooth the surfaces with a smoothing plane.
- Smooth the surfaces with glass paper
- Fill cracks or holes with wood filler
- Rub down the surfaces with fine glass paper or steel wool
- Dust off the surfaces.
- Apply appropriate finish such as wax or varnish

Finishing edges of acrylic

Acrylic sheets usually do require any finish. They already have a good surface finish. The edges, however, will have to be smoothed and polished.

MODULE 5

ENTREPRENEURSHIP

UNIT 5.1 ENTREPRENEURIAL SKILLS

The entrepreneur

Learners should be able to:

describe the entrepreneur.
explain the functions of entrepreneurs.
discuss the importance of entrepreneurs in Botswana.

An entrepreneur is a person who identifies business opportunity, set up a business and run it profitably. The main functions of the entrepreneur are to:

- generate new business ideas, recognize business opportunities and determine profitable business ideas.
- plan the business and identify the resources required.
- bring together and organize resources into a business unit.
- take decision on what to produce, how to produce, where to produce and in what quantity to produce.
- take responsibilities and risks of employing resources of land, labour and capital to produce goods and services.

In Botswana entrepreneurs are important because they help the economy in the following ways:

- As many people become self-employed there is a meaningful drop in unemployment figures.
- Prosperity is spread among many people.
- It is possible that new products and technology will be developed
- Job opportunities can be created.
- There will be fewer crime and welfare problems.

Suggested Teaching and Learning Activities

☞ Ask learners to complete the table below (self assessment).

Entrepreneurial characteristics	Yes	No
Can you think of a business idea which is new and get it started?		
Are you a punctual person?		
Do you stick to your deadlines and keep your promise?		
Are you full of energy and health?		
Are you full of bright ideas?		
Do you like to sit around and do nothing?		
In class, do you chat with friends and disturb other people who are working?		
Are your books in a mess?		

The number of Yes as against No is a fair idea of entrepreneurial qualities of a pupil.

- Let pupils name some business people they know.
- Let them state the characteristics business people usually have.
- Let them suggest some of the characteristics they have themselves.
- Let their friends confirm or deny the characteristics they think they have.
- Let pupils complete the table below (self assessment).

Skills	**1**	**2**	**3**	**4**	**5**
Riding a bike					
Baking a cake					
Keeping a tidy room					
Washing dishes					
Using a calculator					
Typing					
Growing plants					
Drawing and painting					
Playing a musical instrument					
Skipping					
Tying various knots					
Using a computer					
Changing a plug					
Constructing things					
Mending things					
Erecting a tent					
Swimming					
Football					
Dancing					
Singing					
Getting sponsors					
Making friends					
Keeping a secret					
Acting					

1= really good, **2**= quite good, **3**= average, **4**= could be better, **5**= hopeless

The real test is whether their friends would say the same about them.

Using activities;
- Let pupils discuss their strengths and weaknesses
- Analyse the checklist of entrepreneurial characteristics.
- Let them discuss the importance of self-employment.

Support Materials

⇒ A table of 'do you have what it takes'
⇒ An example of Motswana entrepreneur.
⇒ A chart showing small entrepreneurs in different business activities.
⇒ Invite a local entrepreneur to talk to the class

Assessment

✎ Explain the advantages of being self-employed.
✎ List the characteristics of successful entrepreneurs.
✎ Explain the importance of entrepreneurs in Botswana.
✎ Pupils should prepare a personal profile.

Entrepreneurial skills

Learners should be able to:

5.1.2.1 explain the skills and qualities of an entrepreneur.
5.1.2.2 examine the weaknesses of a prospective entrepreneur that can cause business failure.
5.1.2.3 discuss ways of correcting the weaknesses of a prospective entrepreneur.

To be able to start and run your own business successfully, you must have acquired some skills such as planning, marketing, organizing, making decisions, interpersonal, leadership, communication etc. Apart from these general skills, there are some skills on which your business can be based, such as drawing and painting, cooking, playing of musical instruments, mending things, singing, dancing, etc. The entrepreneur should also possess some qualities or characteristics that can make his/her business successful. These qualities or characteristics include hard working, self-confident, focused, adaptable, honest, healthy, independent, perseverance. If the entrepreneur has many of the entrepreneurial qualities and business skills as well as enough money, his or her business is likely to be successful.

Lack of the basic business skills and qualities may lead to business failure. Businesses can also fail because of the following:

- The owner does not know how to run a business
- The owner does not know enough about the activities of the business
- Lack of management skills
- The owner does not keep proper records of transactions
- The business may not be located in the right place
- A disaster such as flood or fire can cause a business failure.

The entrepreneurial qualities or business skills can be learnt, for example in class. There are also business schools which can help to develop entrepreneurial qualities and skills. The government provide assistant to entrepreneurs who have very bright business ideas but do not have money.

Suggested Teaching and Learning Activities

Let learners complete the table below. Think carefully about the following list of skills and **tick** the column which you feel is correct.

Skills	1	2	3	4	5
Washing dishes					
Drawing and painting					
Playing musical instrument					
Constructing things					
Mending things					
Playing football					
Dancing					
Singing					
Acting					
Growing vegetables					
Cooking					
Cleaning homes					

1= really good, **2**= quite good, **3**= average, **4**= could be better, **5**= hopeless

The real test: Would your friends say the same? Why not ask them?

These activities are to test the learners' organization ability. Each learner completes all the activities in stages, starting with activity 1. They would check each organization plan with that of their friends.

Activity 1

You have been asked to organize your **birthday party** this year. You are allowed to have **10** people including yourself. Make a ***list*** of the things you need to think about if your party is to be successful e.g. making a list of people to be invited.

Activity 2

You have to decide upon the menu for the party. **Write** out the menu on a card to be placed on the table.

Activity 3

Can you work out the actual ***cost*** of this menu? You may need to check the prices at the shops.

- Let learners identify institutions which offer help for entrepreneurs from news papers.
- Let learners make a list of do's and don'ts by entrepreneurs.

Support Materials

- A list of skills and entrepreneurial characteristics.
- A list of training institutions for entrepreneurs.
- Newspapers and magazines.

Assessment

- Test of learners' qualities such as **reliability, self-confidence, perseverance** and **initiative**.